

Root cause localization for wind turbines using physics guided multivariate graphical modeling and fault propagation analysis

Chenlong Feng^a, Chao Liu^{a,b,*}, Dongxiang Jiang^{a,c}

^a Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, China

^b Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Tsinghua University, Beijing 100084, China

^c State Key Laboratory of Control and Simulation of Power System and Generation Equipment, Tsinghua University, Beijing 100084, China

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ABSTRACT

Under the background of widespread utilization of wind energy, the improvement of power generation efficiency and the reduction of operation and maintenance costs have increased the demand for anomaly detection and localization of wind turbines through fault propagation analysis, while the large-scale data collected by SCADA systems provide sufficient data support and promising assistance for this purpose. It is an effective and cost-saving solution to this demand to construct a high-precision wind turbine digital twin model under normal operating conditions based on multivariate monitoring data, and to realize root cause localization based on prediction deviation and backtracking analysis. This paper proposes a root cause localization framework for wind turbines based on explicit-implicit knowledge fusion and multivariate time-series graph neural networks, fully utilizing the inter-variable dependencies mining capability of graph structure learning and the temporal prediction advantage of time-series graph neural networks. Five types of explicit knowledge based on explicit rules are constructed to describe the explicit relationships between monitoring variables. The fused knowledge (i.e., graph adjacency matrix) is obtained by fusing the explicit knowledge with the implicit knowledge hidden in the data through a quasi-hard-attention mechanism based on the idea of graph structure learning. A high-precision digital twin model for wind turbines based on fused knowledge is constructed using the multivariate time-series graph neural network. The anomaly degree of monitoring variables' state is defined based on the prediction deviation, with which the propagation of fault's state over time can be accurately described, and the root cause localization through backtracking analysis can be achieved. Case studies are conducted using field fault data from a wind farm, a sugar factory and a thermal power plant. The results show that the proposed framework can provide a solution for studying the fault propagation process and root cause localization, also demonstrating certain robustness and generalizability.

1. Introduction

With the advantages of clean and low-carbon, wind power has made excellent contributions to addressing climate change and environment pollution. In 2022, the global new installed wind power capacity reached 77.6 GW, with a total installed capacity of 906 GW. According to the Global Wind Report 2023, it is expected that the global new installed capacity of onshore wind power will surpass 100 GW for the first time by 2024, and that of offshore wind power will reach a new high of 25 GW by 2025 [1]. With the rapid development of wind power, there is an increasing demand for effective condition monitoring and anomaly detection of wind turbines. Recently, the majority of wind farms are

equipped with supervisory control and data acquisition (SCADA) system, whose collected data has the advantages of large scale, low cost and wide monitoring coverage, making it the most economical choice for implementing intelligent O&M of wind turbines [2,3]. However, this system is currently only used for condition monitoring and alarming, preventing the large amount of stored historical data from creating economic value. In addition, many existing studies focus on anomaly detection for wind turbines based on SCADA data [2–6], but it is difficult to identify the location of the root cause. Therefore, both the industry and academia are eager to realize the root cause localization for wind turbines based on SCADA data to reduce operation and maintenance (O&M) costs and improve economic benefits of wind farms.

* Corresponding author.

E-mail address: cliu5@tsinghua.edu.cn (C. Liu).

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Considering the low sampling rate of SCADA data (typically 10 minutes) [7] and the limited amount of information it contains compared to the vibration data, it is inconvenient to process this data using methods similar to those used for vibration data (e.g., Fourier transform). In most cases, SCADA data describes the normal state of the equipment, and the proportion of abnormal state is small. Consequently, it can be considered to construct a high-precision digital twin model based on multivariate monitoring data for describing the normal operating conditions of the equipment, and then the root cause localization for abnormal state can be implemented through using prediction deviation and backtracking analysis, finally addressing the aforementioned needs. The challenge of this solution lies in how to construct a high-precision digital twin model to describe the normal state of the equipment in a manner that better reflects the actual process. The term “better reflects the actual process” means that the complex system monitored by various sensors can be abstracted as monitoring variables and inter-variable dependencies. In other words, a complex mechanical system can be described by a complex graph structure consisting of monitoring variables (nodes) and relationships between them (edges), and the update of each node’s state depends on capturing historical time-series information and exploring the spatial structure information from the neighboring variables. Therefore, the construction of this digital twin model can be academically summarized as that of a multivariate time-series forecasting model based on graph, which can abstract multivariate time-series data into a combination of intra-variable temporal dependencies and inter-variable spatial dependencies in a manner that better reflects the actual process, and give predictions for all variables.

To achieve high-precision multivariate time-series forecasting, many time-series forecasting models based on typical deep learning methods have made efforts [8–10]. However, these models primarily consider the influence of temporal dependencies on the forecasting results, and pay less attention to the interactions between variables (i.e., inter-variable spatial dependencies) or pay attention to it only implicitly, which results in incomplete information acquired by the models. Graph-structured data, with its explicit node connections, can naturally address this issue. Therefore, on the one hand, multivariate time-series data need to be abstract into graph-structured data, where monitoring variables serve as nodes and the implicit mutual relationships between variables serve as edges. On the other hand, the outstanding ability of graph neural networks in processing graph-structured data needs to be utilized to construct multivariate time-series graph neural networks (MTGNN) to further improve forecasting accuracy [11].

Abstracting multivariate monitoring data into a graph can be achieved from two perspectives: rule-driven (or experience-driven) and data-driven. The rule-driven method relies entirely on physical rules or experts’ experience to describe the inter-variable dependencies to construct a graph based on this explicit knowledge, but it is difficult to implement and the obtained knowledge generally cannot fully and accurately express the relationship between variables. The data-driven method is to learn the inter-variable dependencies from data, which is also known as the graph structure learning (GSL), the reason for which is that most scenarios cannot meet the requirement of “using a pre-defined graph for model training” [12]. Since the information hidden in a large amount of historical monitoring data follows stable statistical patterns, the implicit knowledge obtained from this method can partially and correctly express the dependencies between variables. Therefore, it can be considered to adaptively incorporate different explicit knowledge during the process of obtaining implicit knowledge from data, to jointly learn a more accurate graph structure based on physics & data.

Specifically, to implement the above mentioned scheme of constructing a high-precision digital twin model and realizing root cause localization, this paper proposes a root cause localization framework for wind turbines based on explicit-implicit knowledge fusion and MTGNN by combining the advantages of physics & data-based GSL and MTGNN time-series forecasting. Firstly, the explicit knowledge based on different

rules is designed to fully explore the inter-variable dependencies from the perspective of physics & data, and the explicit and implicit knowledge on wind turbines are fused through a quasi-hard-attention mechanism to abstract the wind turbine into variables and their interactions (i.e., a graph) more accurately. Secondly, a high-precision digital twin model is constructed, namely a MTGNN model based on multivariate time-series data, to accurately and timely describe the normal operating state of the wind turbine, and the anomaly degree of monitoring variables’ state is designed based on this model to accurately describe the nodes’ state. Finally, the propagation of fault’s state over time among different variables (i.e., fault propagation analysis) is obtained, and the root node of the fault can be found through backtracking analysis to realize the root cause localization.

The main contributions of this paper are as follows:

- i) Design explicit knowledge based on different rules, including mechanical transmission relationships, temperature influence relationships, correlation, mutual information and causality, for describing the inter-variable dependencies of wind turbines, and use a quasi-hard-attention mechanism to fuse explicit and implicit knowledge in a GSL framework to obtain an accurate physics & data based graph for wind turbines.
- ii) Use the physics & data based graph and MTGNN to construct a high-precision digital twin model for wind turbines to carry out fault propagation analysis based on prediction deviation using the designed anomaly degree of monitoring variables’ state, and ultimately realize root cause localization through backtracking analysis.
- iii) Verify the effectiveness of the proposed framework using cases from a real-world wind farm, and further verify the robustness and generalizability of the framework in different application scenarios using cases from a sugar factory and a thermal power plant in the real world.

The remaining parts of this paper are organized as follows. In section II, the overview of research progress in GSL, multivariate time-series forecasting and root cause localization is provided. In section III, the preliminaries including causality and MTGNN are introduced. In section IV, the root cause localization framework for wind turbines based on explicit-implicit knowledge fusion and MTGNN is presented. In section V, the datasets, experiment settings and evaluation metrics required for the case studies are described, and the results are presented and discussed. Finally, the conclusions are summarized in section VI.

2. Related Works

2.1. Graph Structure Learning (GSL)

Graph Structure Learning aims at jointly learning an optimized graph structure and its corresponding graph representations [12]. A typical GSL model contains two trainable parts: one is a GNN encoder for receiving a graph as input and generating embedding vectors required for downstream tasks. The second is a structure learner for modeling the connections between nodes in the graph [12]. Generally, the pipeline of GSL includes three stages: graph construction, graph structure modeling and message propagation. Among them, graph structure modeling is the core of GSL, which can model edge connections to refine the initial graph, and can be divided into three categories: metric-based methods, neural networks-based methods and direct methods.

Metric-based methods use different kernel functions to compute the similarity of node features/embeddings to be edge weights. These methods are mainly based on the assumption of network homogeneity, i.e., edges tend to connect similar nodes [13], and refine the graph structure by facilitating intra-class connectivity to produce more compact representations. AGCN is a GSL model based on distance metric learning, which uses generalized Mahalanobis distance to compute the

similarity, and then uses a Gaussian kernel function to refine the initial graph [14]. GRNC [15], GAUG-M [16], and CAGNN [17] use the inner product of two node embeddings (inner product kernel) to model edge weights. Similarly, cosine kernel [18,19], diffusion kernel [20] and multi-kernel fused function [21] can also be used for edge weights modeling.

Neural networks-based methods use more complicated neural networks to model edge weights. For example, the simple neural networks such as a single-layer neural network [22] and multilayer perceptron [23] can be used for GSL. With the rapid development of attention mechanism in recent years, some methods such as GaAN [24], PGN [25] and MAGNA [26] dynamically amplify or attenuate the connections using different attention mechanisms to generate the adjacency matrix. In contrast, instead of considering only first-order adjacencies, Transformer-based methods such as Graphomer [27] and GraphiT [28] consider all nodes as neighbors and encode the graph structure information into the graph transformer, and then weight the edges with attention.

Different from the above methods, direct methods no longer rely on node representations for edge weights modeling, but treat the adjacency matrix as a learnable free variable in GNN training, which provides greater flexibility but presents more difficulties in learning the matrix parameters. Concretely, some studies (e.g., GLNN [29]) use regularizers to optimize the adjacency matrix, while others use a probabilistic approach to construct the adjacency matrix, i.e., assuming that the intrinsic graph structure is sampled from a certain distribution [12]. For example, LDS-GLNN constructs the adjacency matrix by sampling from the Bernoulli distribution [30].

In this paper, inspired by the idea of Regularized Graph Structure Learning (RGSL) [31], the distance between node embeddings is measured using the inner product kernel to construct the initial adjacency matrix, and then the initial graph structure is discretely sampled by the Gumbel Softmax to generate the desired implicit graph structure.

2.2. Multivariate Time-Series Forecasting Based on GNN

Advances in sensing and data stream processing technologies have spawned an explosive growth in time-series data, making it one of the most common data types used to capture and record activities in many domains. The time-series data can be used not only to focus on the past trends but also for many tasks including forecasting, classification, anomaly detection and data imputation [32]. Time-series forecasting, as one of the important topics in time series analysis, aims to predict future values based on the historical observations in the time series. Traditional time-series forecasting is mainly based on the statistical regression models, such as Auto-Regressive Integrated Moving Average (ARIMA), Vector Auto-Regression (VAR) and Vector Auto-Regressive Moving Average (VARMA), etc., which treat the predicted future value as a linear combination of the historical values. Such methods, due to the natural drawbacks of the “linear combination” assumption, are difficult to achieve satisfactory prediction performance when facing complex time-series relationships (e.g., nonlinear and multivariate inter-series relationships). In recent years, deep learning-based methods, including RNNs, CNNs and Transformers, have achieved great success in time-series forecasting with their outstanding ability of nonlinear temporal and spatial feature capturing.

However, considering that multivariate time-series data involves complex interactions across time and variables [32], one of the biggest limitations of the above deep learning methods is its inability to explicitly model the spatial relationships that exist between time-series in non-Euclidean space, which restricts its expressive performance. Therefore, the solution of effectively learning the complexity in these interactions is to use a model constructed by a network or graph, in which variables serve as nodes and the complex relationships between variables serve as edges. In this context, as a powerful tool for learning non-Euclidean data representations, GNN can show great potential in

explicitly and effectively modeling spatio-temporal dependencies in multivariate time-series, thus enhancing forecasting performance.

The GNN-based time-series forecasting models can be methodologically classified into two categories: spatial dependencies modeling and temporal dependencies modeling.

Spatial dependencies, i.e., inter-time series relationships, have a significant impact on the forecasting performance. Current studies model the spatial dependencies from three perspectives: spectral GNNs, spatial GNNs and hybrid models. Spectral GNNs-based models mainly utilize Chebyshev polynomials to approximate graph convolution to model inter-time series dependencies, such as STGCN [33] and StemGNN [34] models. Spatial GNNs-based models model the dependencies using message passing or graph diffusion, which are certain simplifications from the graph perspective compared to spectral GNNs-based methods. For example, DCRNN [35] and GraphWaveNet [36] model time series data by combining graph diffusion layers into GRU or temporal convolution. In contrast, STGCN(1st) (a second version of STGCN [33]) and ST-MetaNet [37] use GCN and GAT to model spatial dependencies to aggregate information neighboring time series. Subsequent works including STSGCN [38], STFGNN [39], MTGNN [11] and ASTTGN [40] are innovated to model spatial-temporal dependencies from different perspectives to enhance learning capability. Constructing a hybrid model by fusing spectral GNNs and spatial GNNs is another line of innovation, e.g., SLCNN [41] captures spatial relations at multiple granularities by employing ChebConv and local message passing as global and local convolution.

Temporal dependencies modeling is another important element of multivariate time-series forecasting, whose primary target is to learn an effective temporal model to accurately capture the dependencies between data points in the time series. Recurrent-based models, convolution-based models, attention-based models and hybrid models can be applied for this purpose. Recurrent-based models are used to understand inter-temporal dependencies in some early methods, such as DCRNN [35] combines graph diffusion with GRU to model the spatial-temporal dependencies, and ST-MetaNet [37] employs two types of GRU to encode historical observations and capture multiple temporal correlations associated with geographic information. Convolution-based models provide a more effective solution for inter-temporal dependencies modeling. As an example, STGCN [33] employs temporal gated convolution integrating 1-D convolution and gated linear units to facilitate model training. MTGNN [11] utilizes multiple kernel sizes to enhance temporal convolution. Recently, an increasing number of researches focus on attention-based models. STGRAT [42] employs multi-head self-attention in its encoder to combine historical observations with its proposed spatial attention mechanism. Hybrid models are another application to model inter-temporal dependencies. ASTGCN [43] and DSTAGNN [44] learn the temporal correlations by utilizing temporal attention and convolution simultaneously, and STGNN* [45] capture the local-global temporal dependencies by merging GRU and Transformer.

2.3. Root Cause Localization

Currently, there are many methods [6,46,47] based on the SCADA data to implement anomaly detection for wind turbines, but few of them focus on the root cause localization. It is an important and challenging task to realize root cause localization for wind turbines based on the multivariate time-series information. However, many existed studies on root cause localization focus on power station circuits [48], analog circuits [49] and reliability evaluation [50], and the ones for wind turbines are relatively few, such as Zhang et.al [51] designed a heterogeneous nodes complex network based on the complex network theory to localize the root cause for wind turbines. The development of multivariate time-series graph neural networks provides a new idea for root cause localization, i.e., given a high-precision model under normal condition based on the multivariate time-series graph neural networks, when the

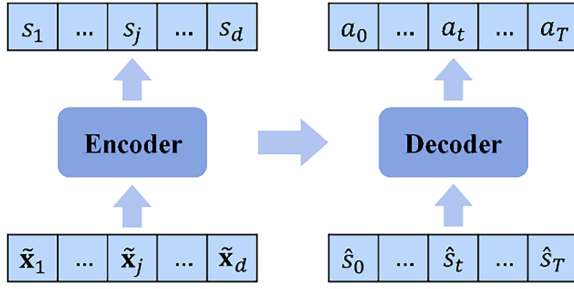


Fig. 1. Policy illustration of CORL model.

fault state propagates through the system, the order of deviation from the normal state for each monitored variable is not consistent, thus root cause localization can be achieved when this order is clarified by backtracking analysis.

3. Preliminaries

3.1. Causality

In many practical applications, identifying the causal structure from monitoring data is an important but challenging task. Wang et al. [52] proposes a causal structure discovery algorithm named Causal discovery Ordering-based Reinforcement Learning (CORL) by taking advantage of the reduced space of variable orderings and the powerful search capability of RL, and verifies that this algorithm outperforms the existing causal discovery algorithms, so this research uses this method to extract the causal structure from monitoring data for wind turbines.

The architecture of CORL consists of an encoder module and a decoder module, and uses RL for neural network optimization, as shown in Fig. 1.

Encoder: It is employed to map the monitoring data to the embedding state space.

Decoder: It is employed to map the state space to the action space.

Optimization: The optimization target of CORL is to learn a policy maximizing the desired cumulative gain.

Variable Selection: For the variable orderings obtained by the above method, CORL utilizes different variable selection methods to get the final causal structure graph.

More details about CORL see Ref. [52,53].

3.2. Multivariate Time-Series Graph Neural Network (MTGNN)

The MTGNN [11] is a general graph neural network framework designed for multivariate time-series data, which consists of graph learning module, graph convolution module, temporal convolution

module and skip connections & output module. However, in this paper, the graph learning module is dropped to use the predefined graph learned from explicit and implicit knowledge. The framework of MTGNN used in this paper is shown in Fig. 2. Firstly, a 1×1 standard convolution is used to project the inputs into a latent space, and then the temporal and spatial dependencies are alternately captured by the temporal convolution modules and graph convolution modules. Residual connections and skip connections are utilized to avoid the gradient vanishing. Finally, the hidden features are projected by the output module according to the desired dimension to produce results.

3.2.1. Graph Convolution Module

Graph convolution module can fuse a node's information with its neighbors' information to capture the spatial dependencies. The module consists of two mix-hop propagation layers to process the information flow passed through each node by the information propagation step and information selection step.

Specifically, given the graph structure A , the information propagation step will recursively pass the node's hidden state $H^{(k-1)}$ of the previous step along with the original state H_{in} , to obtain the processed hidden state $H^{(k)}$. A hyper parameter β is designed to control the ratio of retaining the root node's original state to avoid the hidden state convergence problem caused by the stacking of multilayer graph convolutions. Mathematically, the information propagation step is defined as

$$H^{(k)} = \beta H_{in} + (1 - \beta) \tilde{A} H^{(k-1)} \quad (1)$$

where $\tilde{A} = \tilde{D}^{-1}(A + I)$, and $\tilde{D}_{ii} = 1 + \sum_j A_{ij}$, I is the unit matrix, D is degree matrix, $k \in [1, K]$ is the current layer, K is the depth of propagation.

The information selection step employs a learnable information attention matrix $W^{(k)}$ for filtering the hidden state information to retain the most important information, which is defined as

$$H_{out} = \sum_{i=0}^K H^{(i)} W^{(i)} \quad (2)$$

where H_{out} represents the output hidden state of the current layer, $H^{(0)} = H_{in}$.

3.2.2. Temporal Convolution Module

Temporal convolution module can extract the high-level temporal features by a set of standard dilated 1D convolution filters. The module consists of two dilated inception layers, where the one followed by a tanh activation function serves as a filter, and the other one followed by a sigmoid activation function serves as a gate to control the information

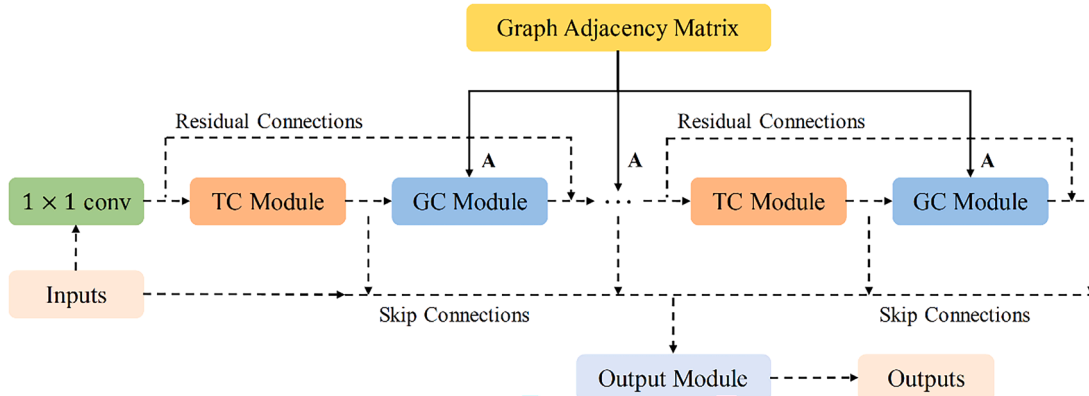


Fig. 2. Framework of MTGNN used in this paper.

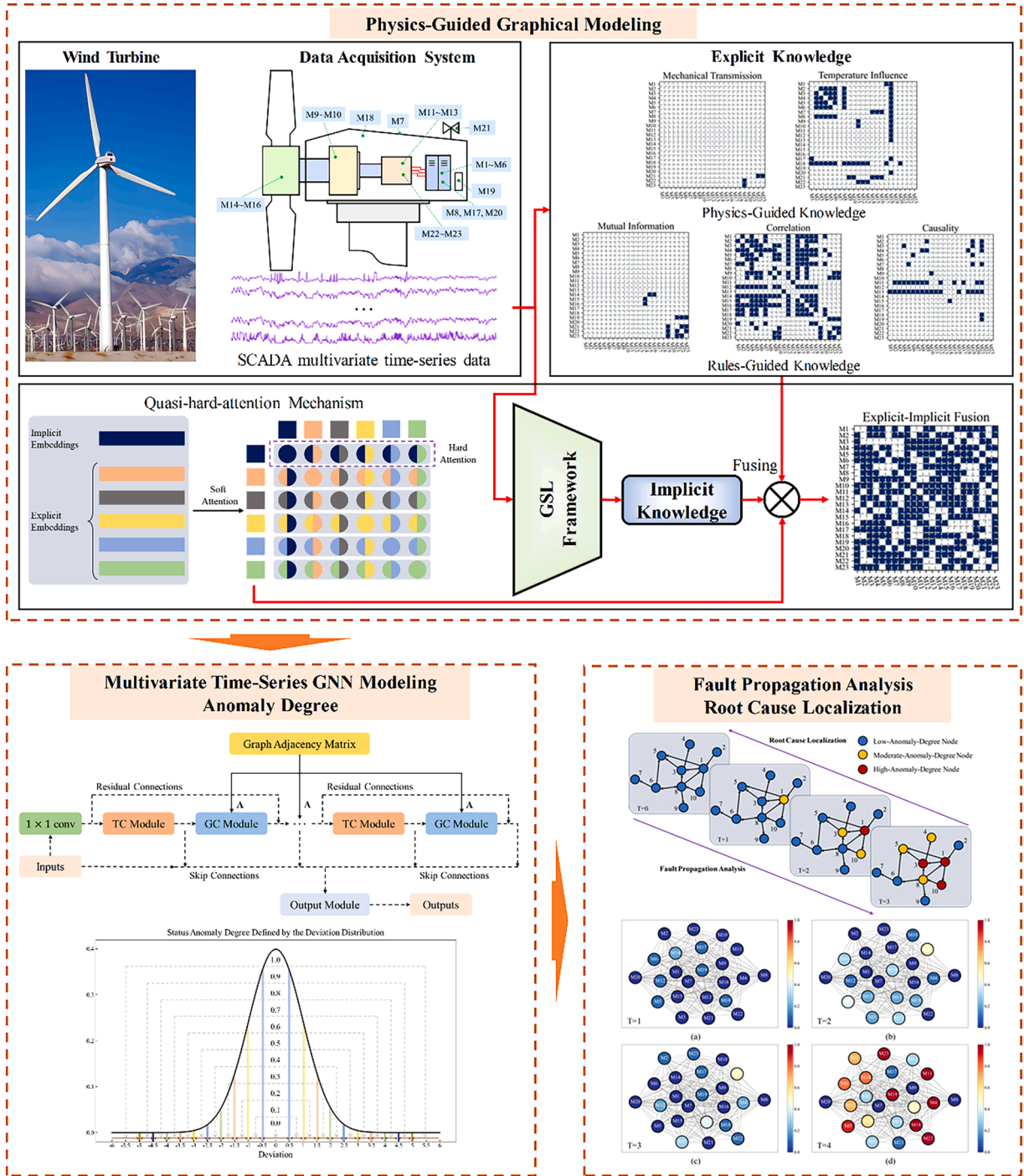


Fig. 3. The overall architecture of the proposed framework for root cause localization.

amount passed to next module.

For the temporal convolution module using 1D convolution filter to extract sequential patterns, the multi-sized filters are beneficial in discovering time-domain patterns at various scales, and the dilated convolution layer is beneficial in reducing model complexity to process long sequences. Thus, four filters of sizes 1×2 , 1×3 , 1×6 and 1×7 are introduced to form the temporal inception layer, and the dilated inception layer combining inception and dilation is proposed. Given a 1D sequence $z \in \mathbf{R}^T$ and the filters consisting of $f_{1 \times 2} \in \mathbf{R}^2$, $f_{1 \times 3} \in \mathbf{R}^3$, $f_{1 \times 6} \in \mathbf{R}^6$ and $f_{1 \times 7} \in \mathbf{R}^7$, the dilated inception layer is defined as

$$z = \text{concat}(z \star f_{1 \times 2}, z \star f_{1 \times 3}, z \star f_{1 \times 6}, z \star f_{1 \times 7}) \quad (3)$$

where $z \star f_{1 \times k}$ denotes the dilated convolution, and its definition is

$$z \star f_{1 \times k}(t) = \sum_{s=0}^{k-1} f_{1 \times k}(s) z(t - d \times s) \quad (4)$$

where d is the dilation factor.

3.2.3. Skip Connection & Output module

Skip connection layers is a standard convolution,

$$h \star f_{1 \times L_i}(t) = \sum_{s=0}^{L_i-1} f_{1 \times L_i}(s) h(t - s) \quad (5)$$

where $\mathbf{h}$ denotes the input of the layer, L_i is the input sequence length of the i^{th} skip connection layer.

Output module consists of two 1×1 standard convolution layers, and can transform the input channel dimension to the desired output dimension. In this paper, the prediction value is the value at next step, so the desired output dimension is 1.

More details about MTGNN see Ref.[11].

4. Methodology

This paper proposes a three-stage root cause localization framework for wind turbines based on explicit-implicit knowledge fusion and multivariate time-series graph neural network, as shown in Fig. 3. In the first stage, five types of explicit knowledge based on mechanical transmission relationships, temperature influence relationships, correlation, mutual information and causality are defined respectively, and the implicit knowledge learned from data is fused with these explicit knowledge using a quasi-hard-attention mechanism in a GSL framework to obtain the explicit-implicit fused knowledge (i.e., the adjacency matrix based on the multivariate monitoring data). In the second stage, a high-precision digital twin model for wind turbines based on MTGNN is constructed using the fused knowledge and the anomaly degree for describing the monitoring variables' state is designed. In the third stage, this anomaly degree is used to obtain the anomaly degree distribution map of the wind turbine at each moment, i.e., the propagation of fault's state over time, and the root cause localization is achieved through backtracking analysis.

4.1. Problem Formulation

In this paper, the construction of a digital twin model for wind turbines can be abstracted as that of the multivariate time-series forecasting model based on MTGNN. Let $\mathbf{z}_t \in \mathbf{R}^N$ denotes the value of an N -dimensional variable at moment t , where $z_t[i] \in \mathbf{R}$ denotes the value of i^{th} variable at moment t . Given an observations sequence of length P on this multivariate variable, $\mathbf{X} = \{\mathbf{z}_{t_1}, \mathbf{z}_{t_2}, \dots, \mathbf{z}_{t_P}\}$, the prediction goal is the value at next moment, $\mathbf{Y} = \{\mathbf{z}_{t_{P+1}}\}$. The goal of model training is to construct a mapping function $f(\cdot)$ from $\mathbf{X}$ to $\mathbf{Y}$ by minimizing the absolute loss.

The graph-related definitions required to construct a MTGNN model in this paper are given as follows:

Definition 4.1. (Graph): The graph is mathematically formulated as $G = (V, E)$, where V is the set of nodes and E is the set of edges. In this paper, the nodes are the monitoring variables of SCADA system for wind turbines and the edges are the interactions between different variables. N is the number of nodes in the graph, i.e., the number of monitoring variables.

Definition 4.2. (Node Neighborhoods): Let $v \in V$ denotes a node, $e = (u, v) \in E$ denotes the edge directed to v from u and then the neighborhoods of node v are defined as $N(v) = \{u \in V | (u, v) \in E\}$.

Definition 4.3. (Adjacency Matrix): The adjacency matrix is a mathematical representation of a graph, denoted as $A \in \mathbf{R}^{N \times N}$, with $A_{ij} = 1$ if $e_{ij} \in E$ and $A_{ij} = 0$ otherwise.

4.2. Physics-Guided Graphical Modeling

The main target of graphical modeling is to abstract a wind turbine into a graph by correctly extract the relationships between different monitoring variables as fully as possible. In this paper, the physics-guided graphical modeling is mainly achieved from two perspectives: explicit knowledge and implicit knowledge. The explicit knowledge is described from five aspects including mechanical transmission

relationships, temperature influence relationships, correlation, mutual information and causality. The acquisition of the implicit knowledge relies on the learning of relationships hidden in the data under the GSL framework. In addition, during the process of implicit knowledge, a quasi-hard-attention mechanism is used to fuse explicit knowledge and implicit knowledge, aiming at learning implicit knowledge as the core supplemented by explicit knowledge with different weights.

4.2.1. Explicit Knowledge

The explicit knowledge is the edges defined by the explicit rules. This paper defines the relationships between variables, i.e., edges, from five aspects including mechanical transmission relationships, temperature influence relationships, correlation, mutual information and causality.

i) Edges Defined by Mechanical Transmission Relationships

This type of edge mainly describes energy transfer relationship between different subcomponents of a wind turbine during the conversion of wind energy into electrical energy. According to the structure of a wind turbine, energy transfer in the drive chain is a unidirectional process, so this type of edge is unidirectional. Specifically, this unidirectional relationship can be one-to-many or many-to-one.

ii) Edges Defined by Temperature Influence Relationships

This type of edge mainly describes the influence of the ambient temperature on the internal temperature of a wind turbine and the influence between the temperature at different locations inside a wind turbine. It is noted that the ambient, as a constant temperature infinite heat source, cannot be influenced by the internal temperature of a wind turbine, so the influence of the ambient temperature on the internal temperature is unidirectional. In contrast, the influence between the temperature at different locations is bidirectional. In addition, the influence between the temperature of the subcomponents at different locations is linked by the nacelle temperature, i.e., there is no direct influence on each other.

iii) Edges Defined by Correlations

This type of edge describes the linear correlation between different variables of a wind turbine using Pearson Correlation Coefficient, with a selection threshold of 0.7.

iv) Edges Defined by Mutual Information

This type of edge describes information-theory based statistical dependencies between different variables of a wind turbine using Mutual Information, with a selection threshold of 0.7.

v Edges Defined by Causality

This type of edge extracts the causal strength between different variables of a wind turbine using CORL algorithm. In the learning process, the data selected by a sliding window with a window size of three months and a step size of one month is used to learn causality between different variables, and finally the desired causal graph is obtained by averaging the results in all window, with a selection threshold of 0.7.

4.2.2. Implicit Knowledge

The implicit knowledge is acquired by the regularized graph generation module [31]. At each iteration, the module samples the adjacency matrix generated from the trainable node embeddings using Gumbel Softmax to discard small-likelihood or redundant edges in a manner similar to Dropout. The above process can be formulated as follows:

Given $E \in \mathbf{R}^{N \times d}$ as the learned node embedding matrix, d as embedding dimension, θ as probability matrix, and $\theta_{ij} \in \theta$ denoting the

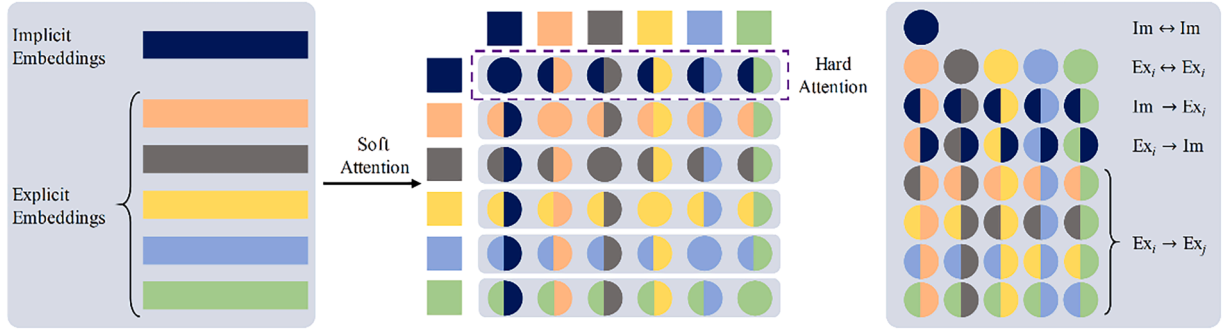


Fig. 4. Explanation for attention weights.

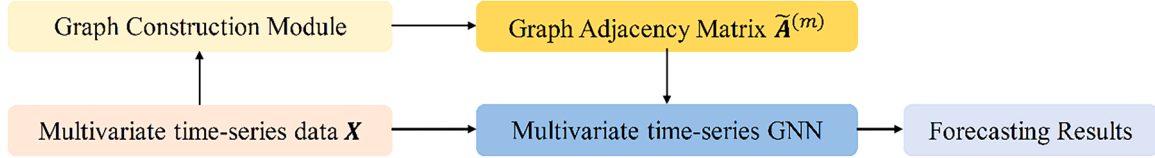


Fig. 5. The concept map of the MTGNN modeling.

probability of retaining edges from time-series i to j , have:

$$\theta = EE^T \quad (6)$$

Given σ as activation function, as s a variable, the sparse adjacency matrix $A^{(l)}$ is defined as:

$$A^{(l)} = \sigma \left(\left(\log(\theta_{ij} / (1 - \theta_{ij})) + (g_{ij}^1 - g_{ij}^2) \right) / s \right) \quad (7)$$

$s.t. \ g_{ij}^1, g_{ij}^2 \sim \text{Gumbel}(0, 1)$

where $A_{ij}^{(l)} = 1$ corresponds to probability θ_{ij} , and 0 corresponds to remaining probability.

Mathematically, Eq. 7 enables the encoding of the inner product between node embeddings to capture the implicit correlations between multivariate time series, which ultimately results in a discrete graph structure, i.e., an implicit graph.

4.2.3. Knowledge Fusion

The explicit and the implicit knowledge describe the interactions between variables in a wind turbine from different aspects, thus the proper fusion of the two will be beneficial for the description of a wind turbine's state. Motivated by the Ref.[31], the Chebyshev polynomial expansion form of graph operations is introduced. Given $X \in \mathbb{R}^{N \times H}$ as the input matrix, $W^{(0)}$, $W_b^{(0)}$, $W^{(l)}$ and $W_b^{(l)}$ as learnable parameters, then the Chebyshev polynomial expansion form of graph operations for $A^{(l)}$ and $A^{(0)}$ are:

$$\tilde{A}^{(0)}(X) = \left(I + D^{-\frac{1}{2}} A^{(0)} D^{-\frac{1}{2}} \right) X W^{(0)} + W_b^{(0)} \quad (8)$$

$$\tilde{A}^{(l)}(X) = \left(I + D^{-\frac{1}{2}} A^{(l)} D^{-\frac{1}{2}} \right) X W^{(l)} + W_b^{(l)} \quad (9)$$

where $I \in \mathbb{R}^{N \times N}$, D is degree matrix.

Inspired by the traditional Mix-up operation and the attention mechanism, the attention module is used to measure the correlation between explicit and implicit knowledge, and the Mix-up operation is used to balance the composition of the two types of knowledge, to fuse the explicit and implicit knowledge.

Therefore, the knowledge fusion relies on the weights between different graph. To facilitate the measurement of these weights, a readout function is firstly introduced to obtain the embedding repre-

sentations $\vec{a}^{(0)}$ and $\vec{a}^{(l)}$ for the implicit and explicit knowledge, and the graph feature representation matrix M is obtained after concatenating all the embedding vectors, which is expressed as:

$$\vec{a}^{(0)}(X) = \mathcal{R}(\tilde{A}^{(0)}(X)) \quad (10)$$

$$\vec{a}^{(l)}(X) = \mathcal{R}(\tilde{A}^{(l)}(X)) \quad (11)$$

$$M = \text{concat}(\vec{a}^{(0)}(X), \vec{a}^{(1)}(X), \dots, \vec{a}^{(l)}(X), \dots, \vec{a}^{(L)}(X)) \quad (12)$$

where $\mathcal{R}(\cdot)$ is readout function, here is the Mean function. $l \in [1, L]$ is the number of the explicit graph, and L is the total number of explicit graphs.

The matrix S of weight coefficients between different knowledge is calculated by the soft attention layer, and the formula is:

$$S = \text{softmax} \left(\frac{(W_q M)(W_k M)^T}{\sqrt{d_k}} \right) \quad (13)$$

where $\text{softmax}(\cdot)$ is the activation function, W_q and W_k are the learnable parameters, d_k is the scaling factor.

The calculated matrix includes five categories of attention weights in the meaning of the implicit knowledge self-attention (denoted as "Im $\leftrightarrow$ Im"), the explicit knowledge self-attention (denoted as "Ex _{i} $\leftrightarrow$ Ex _{i} "), the implicit knowledge's attention to different explicit knowledge (denoted as "Im $\rightarrow$ Ex _{i} "), different explicit knowledge's attention to the implicit knowledge (denoted as "Ex _{i} $\rightarrow$ Im"), the attention between different explicit knowledge (denoted as "Ex _{i} $\rightarrow$ Ex _{j} " ($i \neq j$)), as shown in Fig. 4. In our problem, the implicit knowledge is obtained with the support of a large amount of data and has a regular invariance in the probability sense, while the explicit knowledge is obtained on the basis of different explicit rules and has a variability in the practicability sense. Therefore, the knowledge fusion considered in this paper is implemented with the implicit knowledge as the core and the explicit knowledge as the supplement. Based on the above considerations, the three types of attention weights including "Ex _{i} $\leftrightarrow$ Ex _{i} ", "Ex _{i} $\rightarrow$ Im" and "Ex _{i} $\rightarrow$ Ex _{j} " ($i \neq j$) are ignored, while the attention weight of "Im $\leftrightarrow$ Im" is enforced to be set to 1 and only the attention weight of "Im $\rightarrow$ Ex _{i} " is concerned in a quasi-hard-attention manner. Finally, based on these two types of filtered attention weights, the explicit knowledge $\tilde{A}^{(l)}$ and the learned implicit knowledge

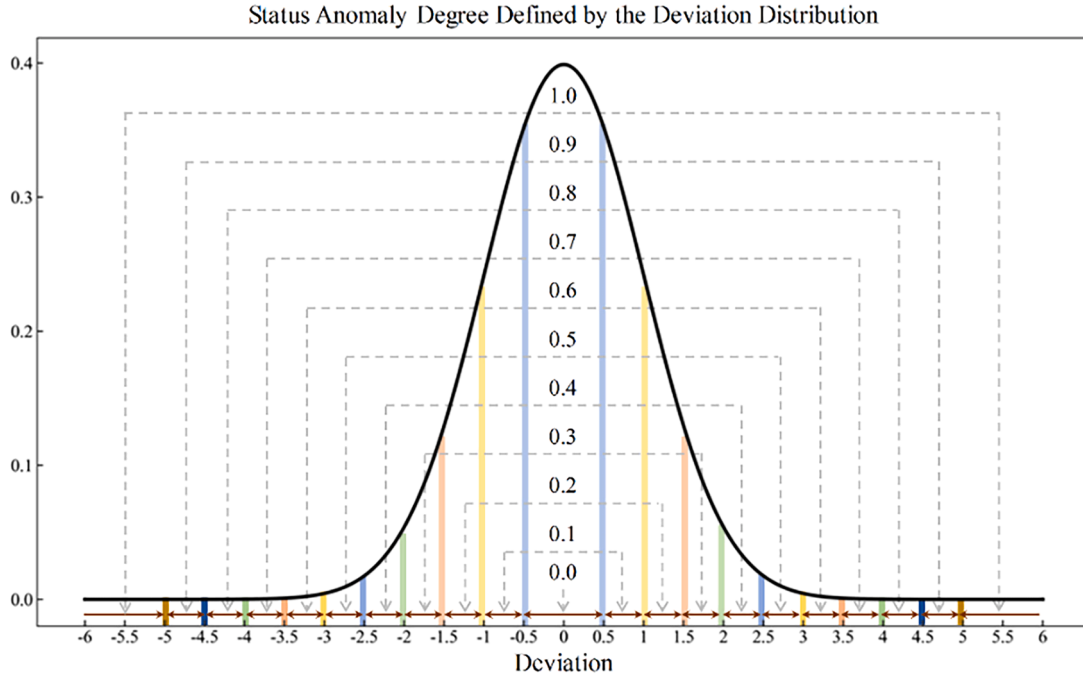


Fig. 6. Definition of anomaly degree.

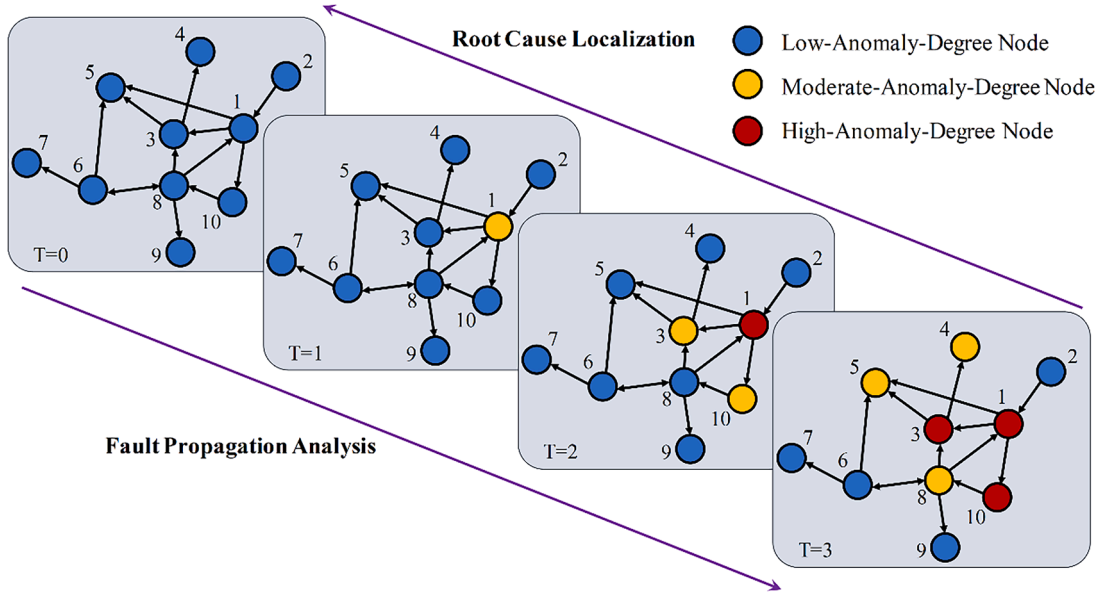


Fig. 7. Fault propagation analysis & root cause localization.

$\tilde{A}^{(0)}$ are fused together, and the formula is:

$$\tilde{A}^{(m)}(X) = \tilde{A}^{(0)}(X) + \sum_{l=1}^L \alpha_l \tilde{A}^{(l)}(X) \quad (14)$$

where α_l is the implicit knowledge's attention to different explicit knowledge (i.e., "Im→Ex_l"), $\tilde{A}^{(m)}$ is the fused knowledge.

4.3. Multivariate Time-Series Graph Neural Network Modeling & Anomaly Degree

This stage aims to construct a high-precision digital twin model based on MTGNN. As shown in Fig. 5, the multivariate time-series data X

is first used to learn the graph adjacency matrix fusing explicit and implicit knowledge based on the graph construction module proposed in Section 4.2. Then the multivariate time-series data X is fed into the MTGNN along with the fused knowledge $\tilde{A}^{(m)}$. After extracting inter-variable dependence and temporal dependence in the data via the graph convolution module and temporal convolution module of the MTGNN respectively, the output module gives the predicted value $\hat{Y}$ at the next moment. Finally, the absolute loss is used to optimize the difference between the target value Y and the predicted value $\hat{Y}$ in order to construct a high-precision model. This training process can be formulated as:

$$\hat{Y} = f(X, \tilde{A}^{(m)}; \Theta) \quad (15)$$

Table 1
23 parameters of a wind turbine and their symbolic representations.

No.	Parameters	Unit	No.	Parameters	Unit
M1	NCC300 temperature	°C	M13	Maximum generator winding temperature	°C
M2	NCC320 temperature	°C	M14	Temperature, pitch motor 1	°C
M3	Machine side inductance temperature	°C	M15	Temperature, pitch motor 2	°C
M4	Grid side inductance temperature	°C	M16	Temperature, pitch motor 3	°C
M5	Machine side semiconductor temperature	°C	M17	Yaw converter temperature	°C
M6	Grid side semiconductor temperature	°C	M18	Nacelle temperature	°C
M7	Ambient temperature	°C	M19	Active power	kW
M8	Filter plate temperature	°C	M20	Nacelle battery temperature	°C
M9	Gearbox bearing temperature	°C	M21	1-second average wind speed	m/s
M10	Gearbox oil temperature	°C	M22	Generator average speed	rpm
M11	Generator bearing temperature in driving direction	°C	M23	Control generator average speed	rpm
M12	Generator bearing temperature in non-driving direction	°C			

$$L = \text{Loss}(\hat{Y}, Y) \quad (16)$$

where $f(\cdot)$ is a mapping function from X to Y , Θ is the learnable parameter of $f(\cdot)$, $\text{Loss}(\cdot)$ is the absolute loss function.

The definition of anomaly degree is derived from the distribution of model predictions' deviations under normal conditions. Here the distribution is assumed to be Gaussian distribution. According to the parameters of Gaussian distribution, i.e., mean and standard deviation, the anomaly degree of a variable's state is defined, as shown in Fig. 6.

4.4. Root Cause Localization & Fault Propagation Analysis

Root cause localization is the exploration of a fault's root source and relies on the fault propagation analysis. By visualizing the change of node's state over time, the change of each monitoring variable's state

will be clarified, and the propagation of the fault's state among different nodes (monitoring variables) will also be determined. Finally, the root cause localization can be achieved by backtracking analysis in the reverse time axis, as shown in Fig. 7.

5. Case Study

5.1. Dataset and Experiment Settings

The SCADA data collected from a wind farm in North China is used for the case study. 50 operation parameters of a wind turbine are monitored in this SCADA system, including 1-second average wind speed, active power, generator average speed, gearbox bearing temperature and gearbox oil temperature, with a sampling interval of 10 minutes. The time span of this dataset is from January 2019 to December 2020. In this research, 23 monitoring parameters were selected for subsequent studies, whose names and symbolic representations are shown in Table 1. In the data preprocessing stage, the SCADA dataset is first cleaned based on standard wind speed power curves, and then processed by the maximum normalization. The dataset is divided into training set, validation set and test set according to the ratio of 0.6, 0.2, 0.2. Fig. 8 shows partially shows the raw data of this dataset.

To demonstrate the advantages of GNNs and the performance improvement by the proposed explicit-implicit knowledge fusion module, 10 benchmark methods are selected for the comparison of modeling accuracy including LSTM [54], BiLSTM [55], CNN-LSTM, Transformer [56], GCRN [57], STGCN [58], GraphLSTM [59], GAT [60,61], NRI [62] and MTGNN [11] used in this paper. The MTGNN has been described in detail in Section 3 of this paper, and the other nine methods are described as follows.

LSTM: Long Short-Term Memory is a variant of recurrent neural networks that can combine short-term memory with long-term memory through gate control and solve the problems of gradient vanishing and gradient explosion in traditional recurrent neural networks.

BiLSTM: Bi-directional Long Short-Term Memory is a variant of Long Short-Term Memory, which combines Forward LSTM and Backward LSTM to better handle input features.

CNN-LSTM: a hybrid model obtained by simply stacking 1D-CNN and LSTM.

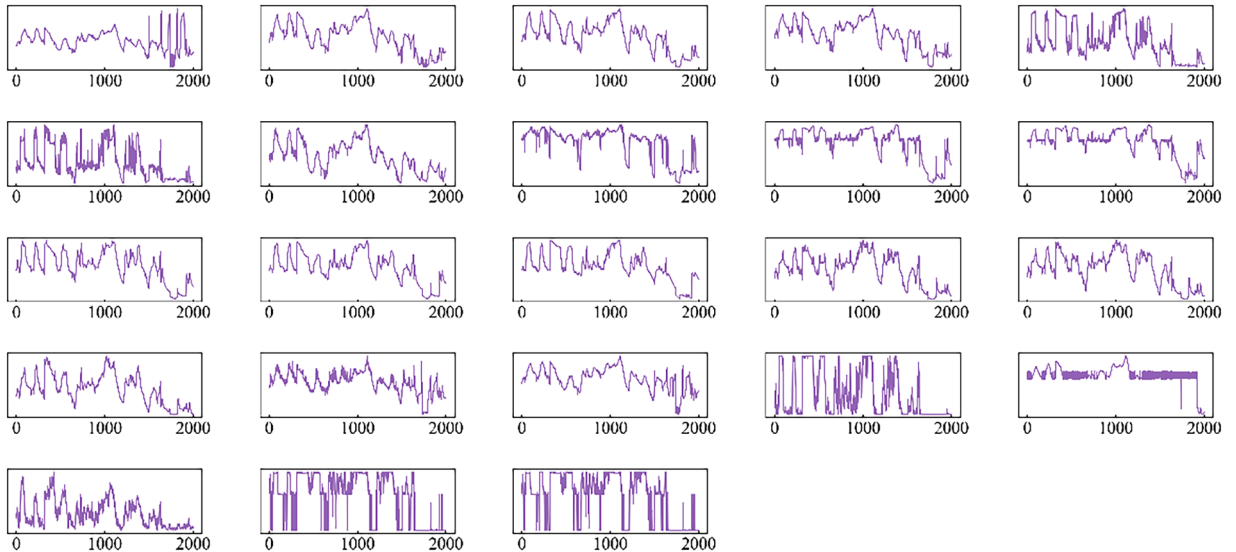


Fig. 8. Raw data of the SCADA dataset, where x-axis is the time (10-mins) and y-axis are the values in the measurements. The units in the y-axis are not shown in the figure for simplicity.

Table 2
Edges Defined by Mechanical Transmission Relationships.

Type of Edges	Edge (u, v)
Conversion of wind energy into mechanical energy.	M21 $\rightarrow$ M22, M21 $\rightarrow$ M23
Conversion of mechanical energy into electrical energy.	M22 $\rightarrow$ M19, M23 $\rightarrow$ M19

Transformer: a neural network that relies entirely on self-attention to compute its input and output representations.

GCRN: As a generalized form of traditional recurrent neural networks over arbitrary structures, Graph Convolutional Recurrent Networks is a deep learning model that can predict structured sequential data.

STGCN: Spatio-Temporal Graph Convolutional Networks is a classical spatio-temporal graph model constructed by the spatio-temporal convolutional blocks, which can automatically learn both spatial and temporal features in data.

DCRNN: Diffusion Convolutional Recurrent Neural Network is a deep learning framework that utilizes the diffusion convolution, the sequence-to-sequence learning frameworks, and the scheduled sampling to capture spatial and temporal dependencies.

GraphLSTM: A neural network that generalizes the LSTM from temporal or multidimensional data to the graph-structured data.

GAT: Graph Attention Network is a graph neural network based on the attention mechanism.

NRI: Neural Relational Inference model is a neural network capable of learning dynamic models in interactive systems in an unsupervised manner to infer hidden interaction structures.

Two metrics including Root Mean Square Error (RMSE) and R-Square (R^2) are used to evaluate the modeling accuracy, where the smaller the RMSE and the larger the R^2 , the higher the prediction accuracy, as shown in Eqs. 17 and 18.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (17)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\bar{y} - y_i)^2} \quad (18)$$

where n is the number of samples, $\hat{y}_i$ is the predicted value at time i , y_i is the monitoring value at the same time, and $\bar{y}$ is the average of samples.

Table 3
Edges Defined by Temperature Influence Relationships.

Type of Edges	Edge (u, v)
Influence of ambient temperature on the temperature of the pitch system and the nacelle.	M7 $\rightarrow$ M14, M7 $\rightarrow$ M15, M7 $\rightarrow$ M16, M7 $\rightarrow$ M18
Interaction between nacelle temperature and internal components temperature.	M18 $\leftrightarrow$ M1, M18 $\leftrightarrow$ M2, M18 $\leftrightarrow$ M3, M18 $\leftrightarrow$ M4, M18 $\leftrightarrow$ M5, M18 $\leftrightarrow$ M6, M18 $\leftrightarrow$ M8, M18 $\leftrightarrow$ M9, M18 $\leftrightarrow$ M10, M18 $\leftrightarrow$ M11, M18 $\leftrightarrow$ M12, M18 $\leftrightarrow$ M13, M18 $\leftrightarrow$ M20, M18 $\leftrightarrow$ M17
Temperature influence on the drive chain.	M21 $\rightarrow$ M14, M21 $\rightarrow$ M15, M21 $\rightarrow$ M16, M21 $\rightarrow$ M9, M21 $\rightarrow$ M10, M9 $\rightarrow$ M11, M10 $\rightarrow$ M11, M22 $\rightarrow$ M11, M22 $\rightarrow$ M12, M22 $\rightarrow$ M13
Temperature influence in the NCC300 control cabinet.	M1 $\rightarrow$ M17
Temperature influence in the NCC320 control cabinet.	M2 $\leftrightarrow$ M3, M2 $\leftrightarrow$ M4, M2 $\leftrightarrow$ M5, M2 $\leftrightarrow$ M6, M2 $\leftrightarrow$ M8, M3 $\leftrightarrow$ M4, M3 $\leftrightarrow$ M5, M3 $\leftrightarrow$ M6, M3 $\leftrightarrow$ M8, M4 $\leftrightarrow$ M5, M4 $\leftrightarrow$ M6, M4 $\leftrightarrow$ M8, M5 $\leftrightarrow$ M6, M5 $\leftrightarrow$ M8, M6 $\leftrightarrow$ M8

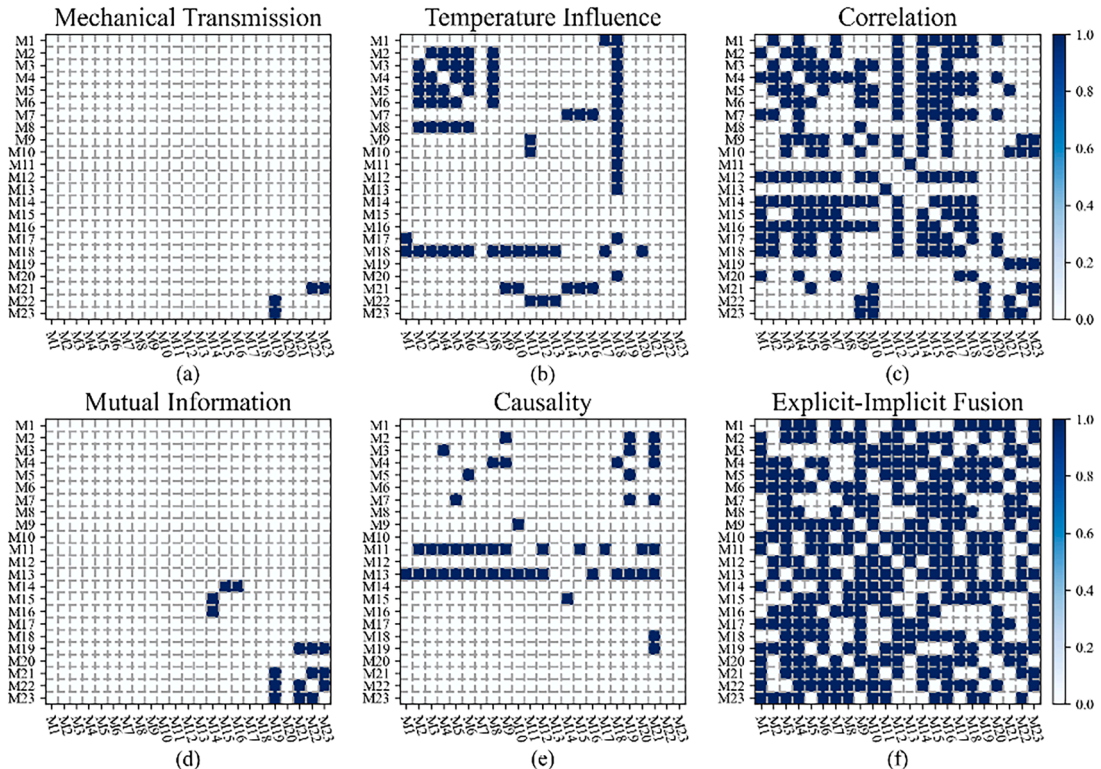


Fig. 9. Explicit Graphs and Fused Graph.

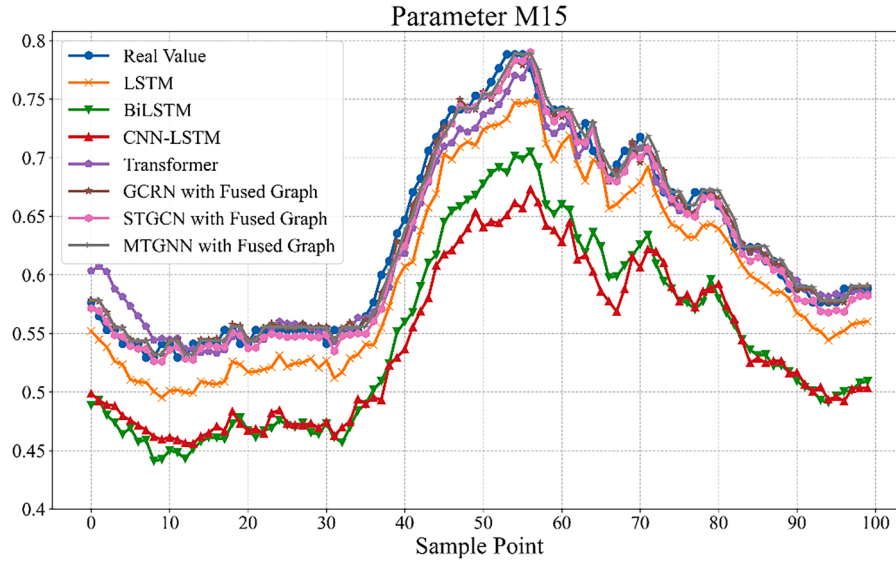


Fig. 10. Forecasting Results of Different Models for Parameter M15.

Table 4

The Modeling Accuracies of 10 Models in 2 Evaluation Metrics (RMSE and R^2).

Model	RMSE							R^2						
	No Graph	G_1	G_2	G_3	G_4	G_5	G_{fused}	No Graph	G_1	G_2	G_3	G_4	G_5	G_{fused}
LSTM	0.0731	—	—	—	—	—	—	0.8346	—	—	—	—	—	—
BiLSTM	0.0752	—	—	—	—	—	—	0.8485	—	—	—	—	—	—
CNN-LSTM	0.0772	—	—	—	—	—	—	0.8330	—	—	—	—	—	—
Transformer	0.0699	—	—	—	—	—	—	0.8224	—	—	—	—	—	—
GraphLSTM	—	0.0631	0.0641	0.0632	0.0636	0.0635	0.0629	—	0.9192	0.9143	0.9096	0.9163	0.9164	0.9145
GAT	—	0.0970	0.1194	0.1556	0.0899	0.0964	0.1026	—	0.7464	0.5942	0.3547	0.7839	0.7378	0.7271
NRI	—	0.0417	0.0416	0.0378	0.0416	0.0417	0.0416	—	0.8569	0.9623	0.9469	0.8825	0.9605	0.9631
GCRN	—	0.0443	0.0479	0.0463	0.0496	0.0460	0.0459	—	0.9647	0.9517	0.9459	0.9518	0.9557	0.9305
STGCN	—	0.0697	0.0591	0.0624	0.0679	0.0694	0.0600	—	0.8687	0.9188	0.9016	0.8793	0.8781	0.9136
MTGNN	0.0525	0.0519	0.0530	0.0541	0.0516	0.0438	0.0427	0.9440	0.9440	0.9428	0.9406	0.9458	0.9655	0.9666

5.2. Results

5.2.1. Study for Graphs

i) Explicit Graph

The mechanical transmission relationships (G_1) in a wind turbine involves some variables such as wind speed, generator speed and active power, which are used to describe the process of converting wind energy into mechanical and electrical energy sequentially after it has been captured by a wind turbine. The connections of the directed edges are shown in Table 2 and Fig. 9 (a).

The temperature influence relationships (G_2) in a wind turbine involves many temperature-based variables, which are used to describe the temperature interactions between variables. The connections of the directed edges are shown in Table 3 and Fig. 9 (b).

The graphs based on correlation (G_3), mutual information (G_4) and causality (G_5) are obtained by the method illustrated in Section 4.2.1, as shown in Fig. 9 (c), Fig. 10 (d) and Fig. 9 (e).

i) Fused Graph

Using the algorithm introduced in Section 4.2.3, the fused graph (G_{fused}) is produced with the implicit knowledge as the core and the explicit knowledge with different weights as the supplement, as shown in Fig. 9 (f). Compared with the proposed five explicit graphs, the fused graph contains richer information, which reflects the limitation and

incompleteness of the explicit knowledge, in other words, there are still many implicit edge information in the data waiting to be mined.

It is worth noting that some edge information in the fused graph is interpretable, e.g., the edge of “M19 $\rightarrow$ M21” expresses the relationship between wind speed and power, which is not directly defined in the explicit knowledge, but is successfully learned from data after mining the implicit knowledge. The edge of “M9 $\leftrightarrow$ M10” expresses the interaction between gearbox bearing temperature and gearbox oil temperature, which is also not directly defined, but is still learned by the proposed algorithm. The learning of these typical relationships provides strong evidence for the effectiveness of the knowledge fusion, and supports the subsequent construction of the digital twin model for a wind turbine.

5.2.2. Modeling Accuracy

The modeling accuracies of seven benchmark models are compared using two evaluation metrics listed in Section 5.1, where the configurations of the input and output parameters are the same for the different models. The time series forecasting results of the different models (traditional time-series neural networks and time-series GNNs) under seven conditions including no graph, five explicit graphs and the fused graph are shown in Table 4. The results under the conditions of no graph and the fused graph are shown in Fig. 10. It can be seen that the time-series GNNs constructed with graph and time-series data has higher accuracy compared to the time-series neural networks constructed with only time-series data. This can be explained that time-series GNNs incorporate the interactions between variables into the training process

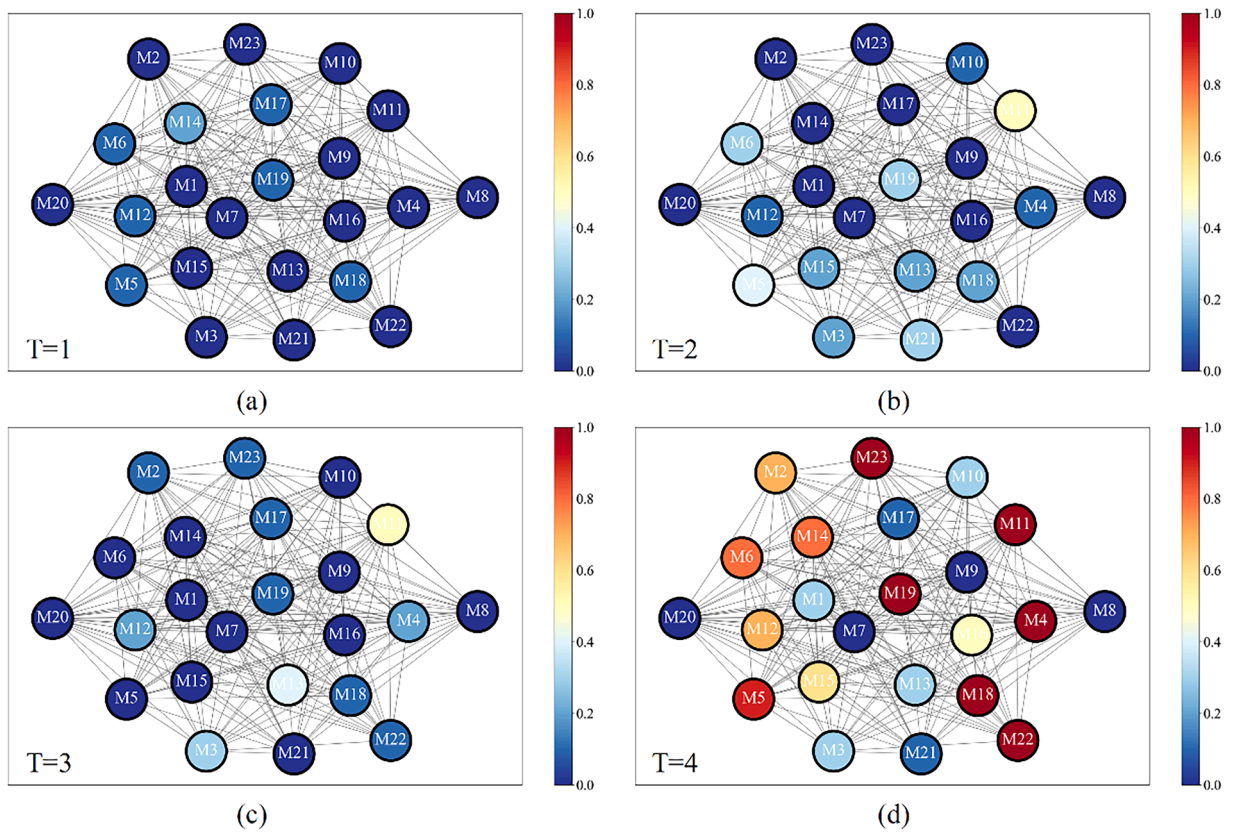


Fig. 11. Fault propagation analysis for Fault "The active power deviation greater than or equal to 500 kW".

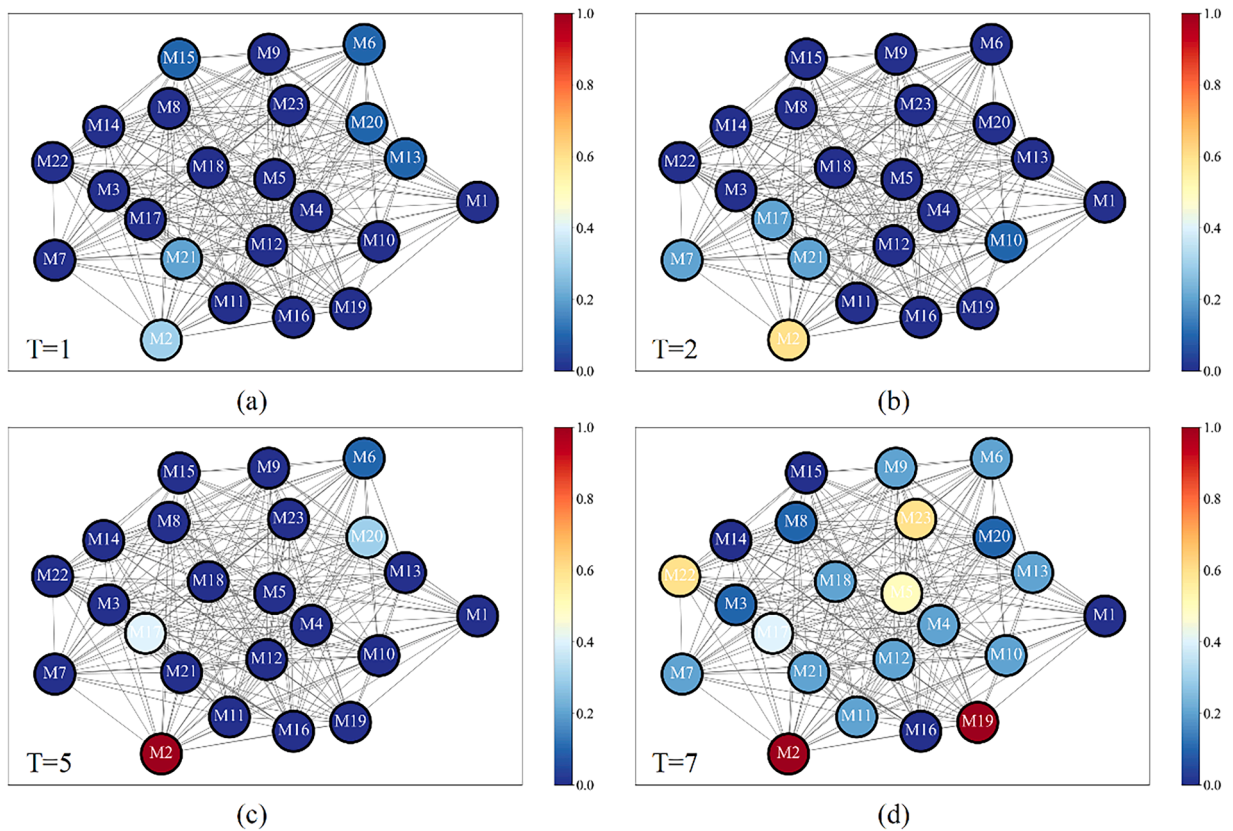


Fig. 12. Fault propagation analysis for Fault "High temperature in two of the three control cabinets".

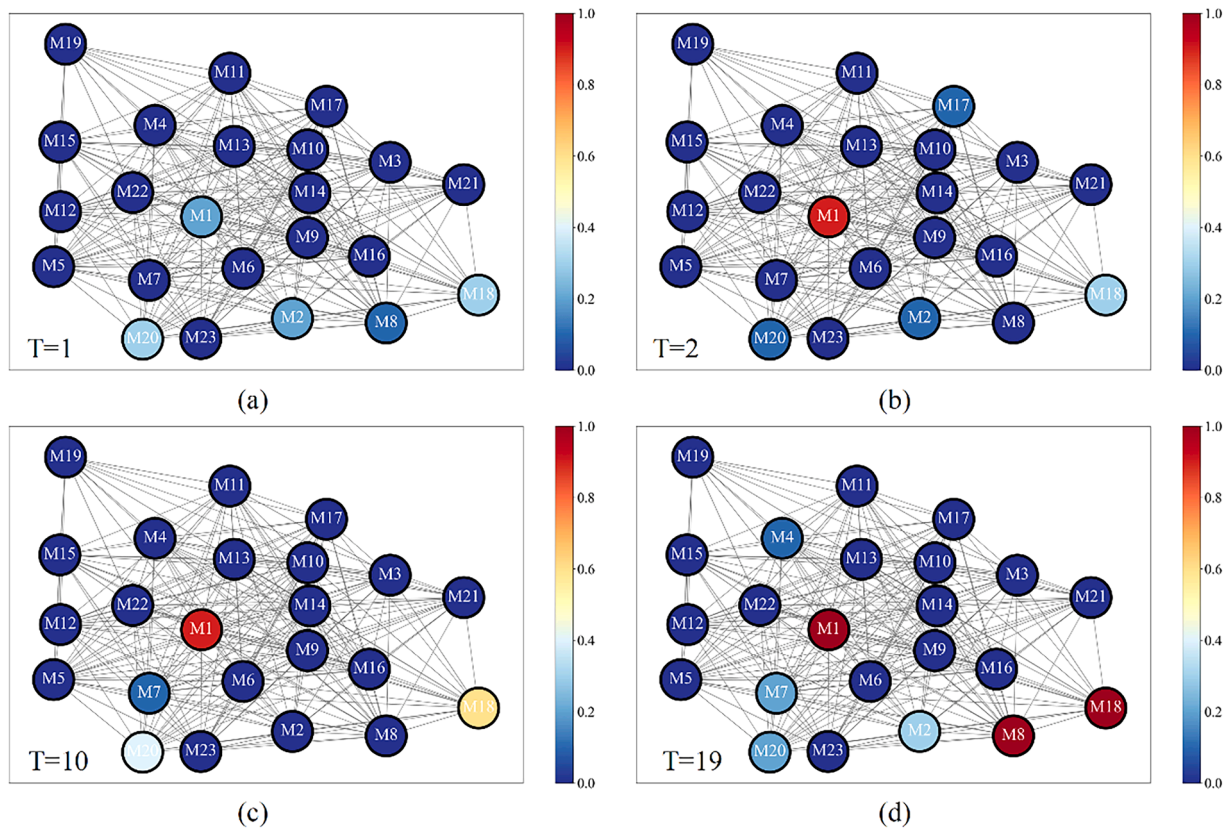


Fig. 13. Fault propagation analysis for Fault “Filter plate temperature exceeding 100 degrees”.

Table 5

32 parameters of Lublin Sugar Factory and their symbolic representations.

No.	Parameters	Unit	No.	Parameters	Unit
N1	P1 – juice pressure (valve inlet)	kPa	N17	P1 – juice pressure (valve inlet)	kPa
N2	P2 - juice pressure (valve outlet)	kPa	N18	P2 - juice pressure (valve outlet)	kPa
N3	T - juice temperature (valve inlet)	°C	N19	T - juice temperature (valve inlet)	°C
N4	F - juice flow (1st evaporator inlet)	m ³ /h	N20	PV - process value (juice flow, 5th evaporator outlet)	m ³ /h
N5	CV-control value (controller output)	%	N21	CV – control value (controller output)	%
N6	X-servomotor rod displacement	%	N22	X – servomotor rod displacement	%
N7	PV-process value (juice level in 1th evaporator)	%	N23	P1 – water pressure (valve inlet)	kPa
N8	Juice temperature (1st evaporator inlet)	°C	N24	P2 – water pressure (valve outlet)	kPa
N9	Juice temperature (1st evaporator outlet)	°C	N25	T - water temperature (valve outlet)	°C
N10	Juice density (1st evaporator inlet)	BX	N26	F – water flow (steam boiler inlet)	t/h
N11	Juice density (1st evaporator outlet)	BX	N27	CV – control value (controller output)	%
N12	Steam flow	t/h	N28	X – servomotor rod displacement	%
N13	Steam pressure	kPa	N29	PV - process value (water level in steam boiler)	%
N14	Steam temperature	°C	N30	Steam flow (steam boiler outlet)	t/h
N15	Vapour pressure	kPa	N31	Steam pressure (steam boiler outlet)	kPa
N16	Vapour temperature	°C	N32	Steam temperature (steam boiler outlet)	°C

in a more explicit form, and the network will be more likely to be trained in a correct direction; while time-series neural networks implicitly consider the interactions between variables into the training process, where the entirely parameterized learning for the interactions has a greater degree of freedom and is more difficult to learn, making the modeling accuracy is hard to surpass that in time-series GNNs.

In addition, the improvement of modeling accuracy by the fused knowledge compared to the explicit knowledge is also a concern. Based on the modeling approach described in Section 4.3, the five explicit graphs and the fused graph are sent to the models for training with the same parameter configurations respectively. As can be seen in Table 4, the models constructed based on the fused graph has the highest (or the nearly highest) accuracy (i.e. minimum RMSE and maximum R^2), which means that the fused graph helps these models to describe the normal state of a wind turbine more accurately. This can be explained that the explicit graphs can only provide partially valid information for model training, while the fused graph is more comprehensive in providing information about the interactions between variables.

5.2.3. Fault Propagation Analysis and Root Cause Localization

In this section, three fault cases from real wind farms are used to verify the effectiveness of the proposed framework: i) the active power deviation greater than or equal to 500 kW, ii) high temperature in two of the three control cabinets, and 3) filter plate temperature exceeding 100 degrees.

i) The active power deviation greater than or equal to 500 kW

The fault is related to the parameter M19 “Active power”, as shown in Fig. 11 (d), when the alarm is raised, the anomaly degree of parameter M19 has reached the maximum value of 1, so the alarm of this fault is reasonable. When looking back at the propagation of the fault, it can be found that in the half hour before the fault occurred, only the parameter

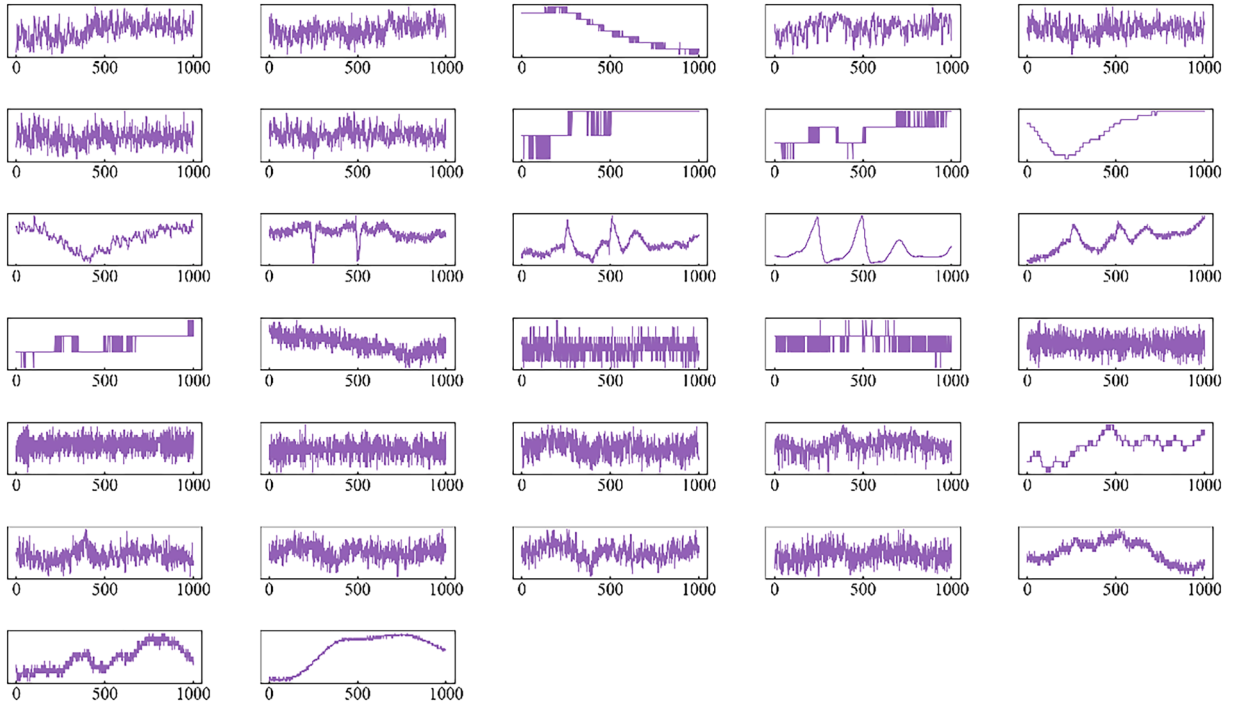


Fig. 14. Raw data of the Lublin Sugar Factory dataset, where x-axis is the time (1-seconds) and y-axis are the values in the measurements. The units in the y-axis are not shown in the figure for simplicity.

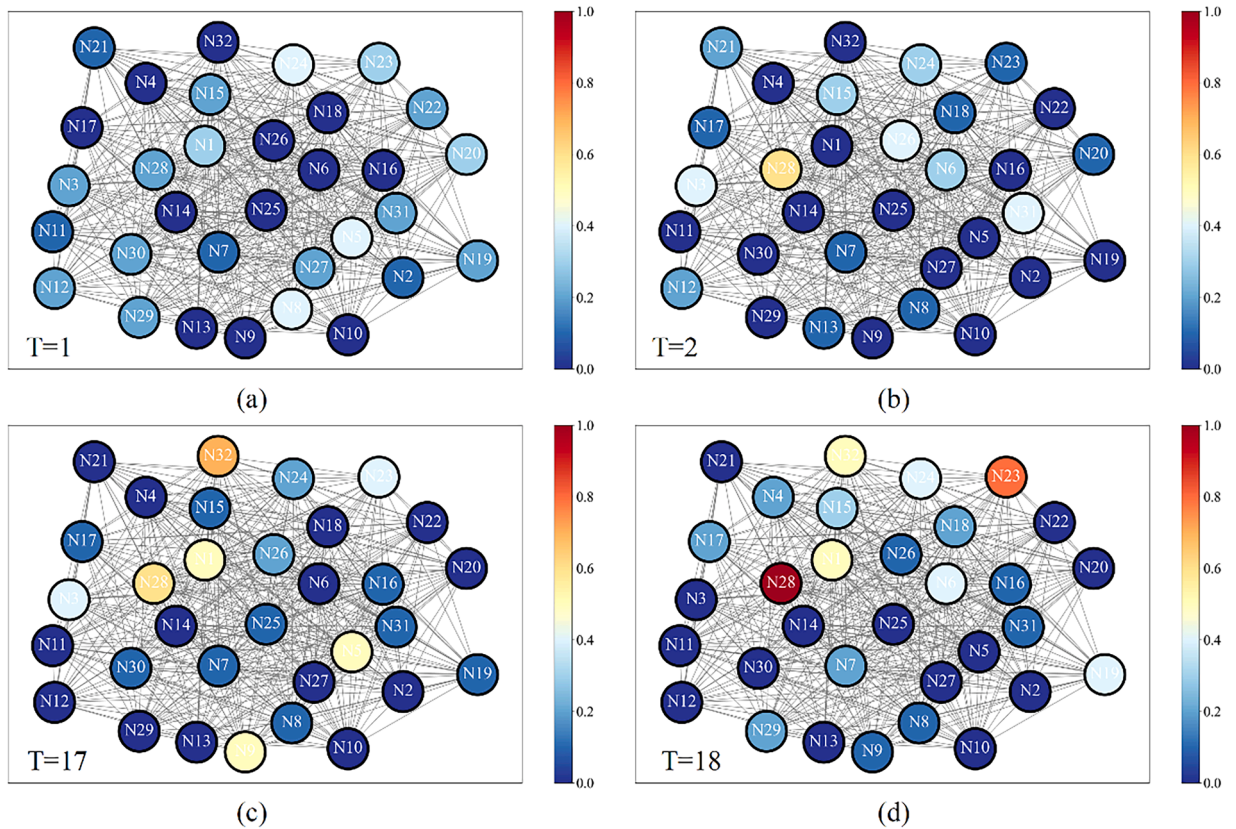


Fig. 15. Fault propagation analysis for Fault "Positioner supply pressure drop on Actuator 3".

M11 "Generator bearing temperature in driving direction" anomaly degree is gradually increasing, until the moment of the fault, the anomaly caused by M11 is expanded to other parameters along the

system network, ultimately causing an anomaly alarm of M19. Therefore, we can locate the root cause of the fault in this case as a generator bearing anomaly.

Table 6
30 parameters of Coal Pulverizer and their symbolic representations.

No.	Parameters	Unit	No.	Parameters	Unit
V1	Coal pulverizer current	A	V16	Coal pulverizer motor phase B stator winding temperature 2	°C
V2	Coal pulverizer lubricating oil pump current	A	V17	Coal pulverizer motor phase C stator winding temperature 1	°C
V3	Coal pulverizer outlet pressure 1	kPa	V18	Coal pulverizer motor phase C stator winding temperature 2	°C
V4	Coal pulverizer outlet pressure 2	kPa	V19	Coal pulverizer motor shaft end bearing temperature	°C
V5	Coal pulverizer outlet temperature 1	°C	V20	Coal pulverizer motor non-shaft end bearing temperature	°C
V6	Coal pulverizer outlet temperature 2	°C	V21	Coal pulverizer grinding bowl upper/lower differential pressure	kPa
V7	Coal pulverizer planetary gearbox bearing temperature 1	°C	V22	Coal pulverizer seal air and primary air differential pressure	%
V8	Coal pulverizer planetary gearbox bearing temperature 2	°C	V23	Coal pulverizer cold air damper opening feedback	%
V9	Coal pulverizer planetary gearbox bearing temperature 3	°C	V24	Coal pulverizer hot air damper opening feedback	%
V10	Coal pulverizer planetary gearbox bearing temperature 4	°C	V25	Coal pulverizer primary air flow	km ³ /h
V11	Coal pulverizer lubricating oil tank temperature	°C	V26	Coal pulverizer primary air flow setpoint	km ³ /h
V12	Coal pulverizer lubricating oil temperature	°C	V27	Coal pulverizer outlet temperature setpoint	°C
V13	Coal pulverizer motor phase A stator winding temperature 1	°C	V28	Coal pulverizer inlet primary air flow	km ³ /h
V14	Coal pulverizer motor phase A stator winding temperature 2	°C	V29	Coal pulverizer inlet primary air temperature	°C
V15	Coal pulverizer motor phase B stator winding temperature 1	°C	V30	Coal pulverizer primary air flow	km ³ /h

ii) High temperature in two of the three control cabinets

The fault is related to parameter M1 “NCC300 temperature” or M2 “NCC320 temperature”, so it is necessary to further confirm the source node of the fault. As shown in Fig. 12 (d), when the fault alarm is raised, the anomaly degree of M2 has reached the maximum value of 1, which is consistent with the fault description. When looking back at the propagation of the fault, it can be seen that in an hour before the fault, only the anomaly degree of M2 is in the gradual increase, and then the anomaly caused by M2 is expanded to other parameters and caused the alarm finally. Therefore, we can locate the root cause of the fault as a temperature anomaly in the control cabinet “NCC320”.

iii) Filter plate temperature exceeding 100 degrees

The fault is related to parameter M8 “Filter plate temperature”. As shown in Fig. 13 (d), when the fault alarm is raised, the anomaly degree of M8 has reached the maximum value of 1, which is consistent with the fault description. When tracing back the propagation of the fault, it can be seen that the anomaly caused by M1 “NCC300 temperature” is firstly extended to M18 “Nacelle temperature”, and then to M8, triggering the alarm finally. Therefore, we can locate the root cause of the fault as a temperature anomaly in the control cabinet “NCC300”.

5.3. Case Studies on Lublin Sugar Factory & Coal Pulverizer & Another Wind Turbine

The proposed framework based on the prediction deviations-based anomaly detection strategy is highly generalizable due to its simplicity and reliability. To demonstrate this point, fault samples from Lublin Sugar Factory public dataset [63], the coal pulverizer monitoring dataset of a thermal power plant and a wind turbine SCADA dataset of another wind farm in the real world are used as case studies in this paper. To reduce the difficulty of modeling and further improve the generalizability of the framework, here only the correlation, mutual information and causation are viewed as the explicit knowledge to fuse with the implicit knowledge, and then a high-precision digital twin model for the Lublin Sugar Factory system, the coal pulverizer or the wind turbine is constructed to carry out the root cause localization of the specific case.

5.3.1. Case from the Lublin Sugar Factory

The data from the Lublin Sugar factory provided by DAMADICS benchmark [63] consists of 32 measurements with a sampling rate of 1 Hz and spans the period from October 29 to November 22, 2001. The symbolic representation of the 32 parameters and their names are shown in Table 5. The methods of data preprocessing and dataset segmentation are the same as ones used for the SCADA data of wind turbines. Fig. 14 shows partially shows the raw data of this dataset.

The artificial fault “Positioner supply pressure drop on Actuator 3” is used for the case study. According to the Data Description File [64], this artificial fault was introduced on November 17, 2001, and reflected in the fact that the parameter N28 “X – servomotor rod displacement” started to decrease at 20 seconds before the fault, deviating from the normal operating condition. Based on the proposed framework, the propagation of parameters’ state in the sugar factory system before the fault is successfully evaluated, as shown in Fig. 15. It can be seen that the state of parameter N28 becomes significantly abnormal at 16 seconds before the fault (i.e., Fig. 15 (b)), and this anomaly then propagates to other nodes, ultimately reaching the maximum anomaly degree at the moment of the fault (i.e., Fig. 15 (d)). In terms of root cause localization, backtracking the above fault propagation, it can be judged that the root cause of this artificial fault from the sugar factory system is the parameter N28 anomaly, which is consistent with the fact and proves that the framework is able to achieve root cause localization.

5.3.2. Case from the Coal Pulverizer

The coal pulverizer data provided by a thermal power plant in eastern China consists of 30 measurements with a sampling rate of 1/30 Hz and spans the period from July 2021 to March 2022. The symbolic representation of the 30 parameters and their names are shown in Table 6. The methods of data preprocessing and dataset segmentation are the same as ones used for the SCADA data of wind turbines. Fig. 16 shows partially shows the raw data of this dataset.

The real fault “Pulverizer Roller Bearing Fault” is used for the case study. According to the pulverizer structure, this fault causes anomalies in the temperature measurements of the pulverizer gearboxes and motors, e.g., V7-V10, V13-V20, etc. Based on the proposed framework, the propagation of parameters’ state in the coal pulverizer before the fault is successfully evaluated, as shown in Fig. 17, where it can be seen that the parameters V8, V10, V13-V18, V20, and V29 exhibit a high anomaly degree at the moment of the fault (i.e., Fig. 17 (d)), which is consistent with the fact. After backtracking the propagation of the fault, the above parameters all reflect the trend of gradually increasing the anomaly degree with time, in which the state of parameter V20 “Coal pulverizer motor non-shaft end bearing temperature” is the earliest to become abnormal (i.e., Fig. 17 (a)). Therefore, it can be judged that the root cause of this fault is the parameter V20 anomaly, and the components related to V20 should be focused on firstly in the actual overhaul.

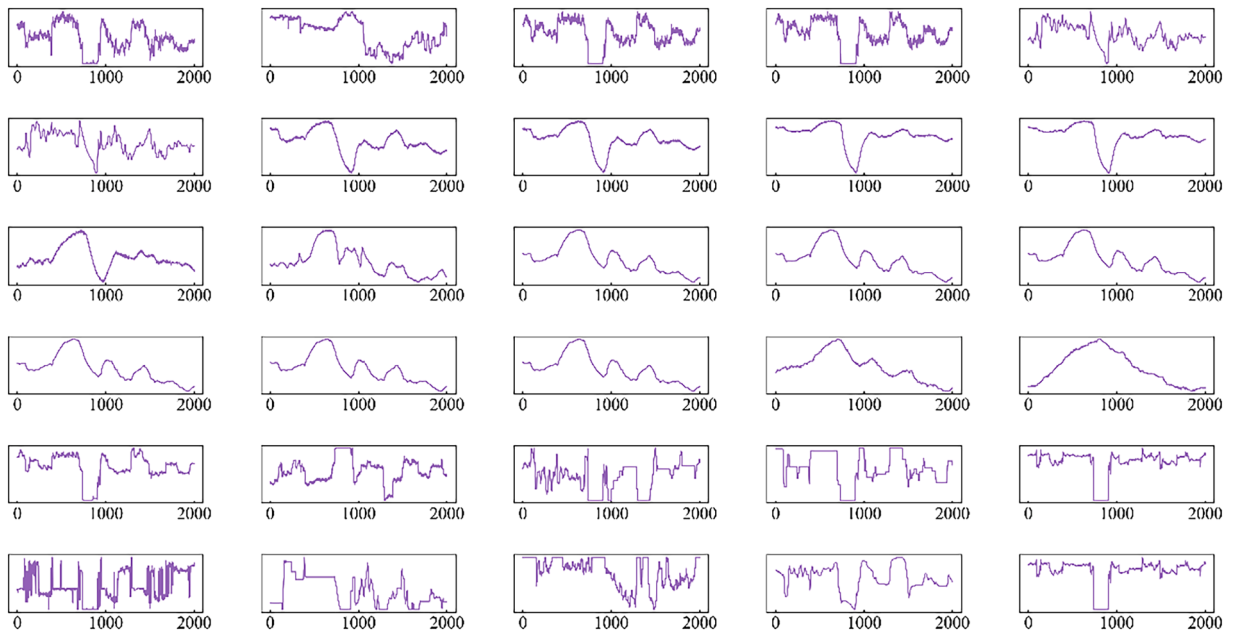


Fig. 16. Raw data of the Coal Pulverizer dataset, where x-axis is the time (30-seconds) and y-axis are the values in the measurements. The units in the y-axis are not shown in the figure for simplicity.

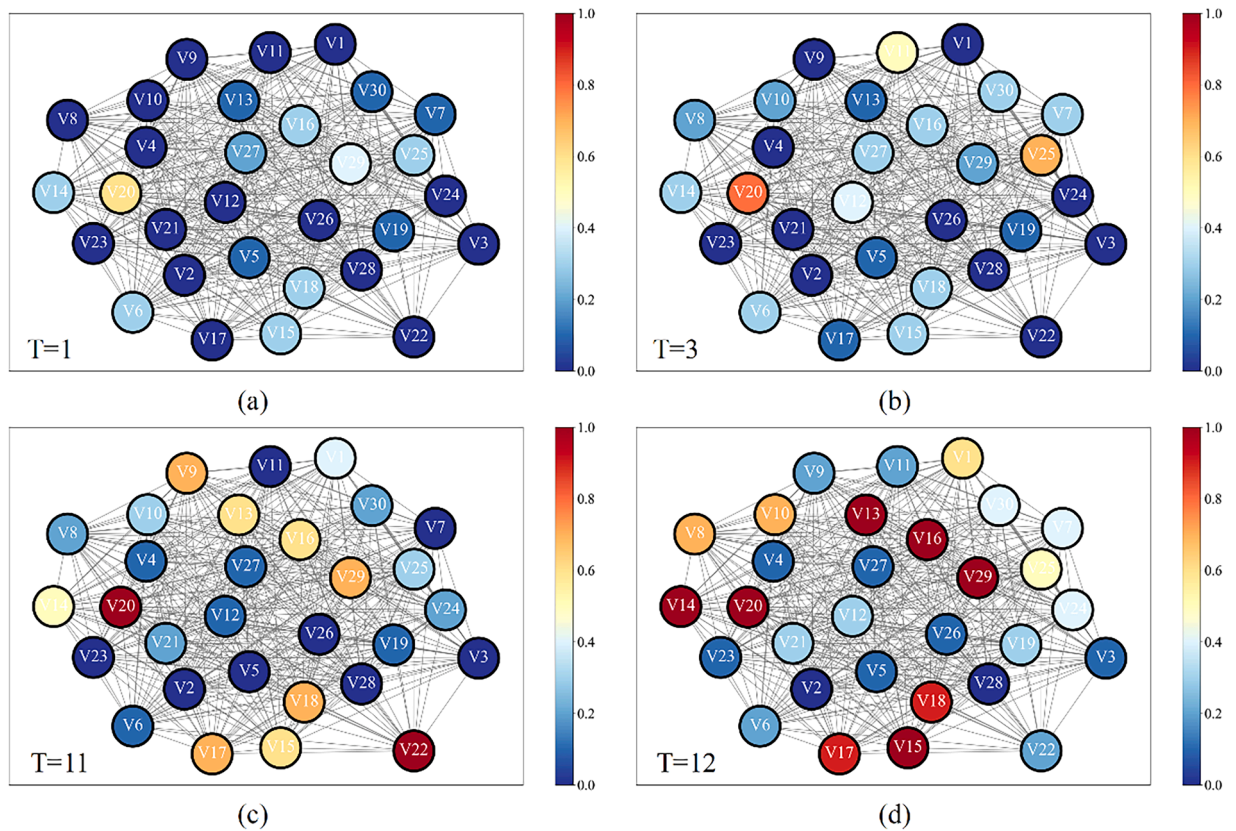


Fig. 17. Fault propagation analysis for Fault “Pulverizer Roller Bearing Fault”.

5.3.3. Case of a wind turbine from another wind farm

The monitoring data of this case provided by a wind farm (different from the aforementioned wind farm) in northeastern China consists of 50 measurements with a 10-minutes sampling interval and spans the period from January 2019 to December 2021. The symbolic representation of the 50 parameters and their names are shown in Table 7.

The real fault “Inverter Abnormality” is used for the case study. According to the location of the measurement sensors, the fault can be related to the monitored parameters in the inverter control cabinet, such as M1-M6, M9, M10, etc. Based on the proposed framework, the propagation of parameters’ state in the wind turbine before this fault is successfully evaluated, as shown in Fig. 18. It can be seen that when the

Table 7
50 parameters of a Wind Turbine and their symbolic representations.

No.	Parameters	Unit	No.	Parameters	Unit
M1	NCC300 temperature	°C	M26	Nacelle battery Voltage	V
M2	NCC320 temperature	°C	M27	Nacelle battery temperature	°C
M3	Machine side inductance temperature	°C	M28	Tower oscillation in driving direction	%
M4	Grid side inductance temperature	°C	M29	Tower oscillation in non-driving direction	%
M5	Machine side semiconductor temperature	°C	M30	1-second average wind speed	m/s
M6	Grid side semiconductor temperature	°C	M31	1-second minimum wind speed	m/s
M7	Amplitude of drive chain oscillation	%	M32	1-second maximum wind speed	m/s
M8	Ambient temperature	°C	M33	Wind direction	°
M9	Maximum fault current	A	M34	Average nacelle position	°
M10	Filter plate temperature	°C	M35	Minimum nacelle position	°
M11	Gearbox bearing temperature	°C	M36	Maximum nacelle position	°
M12	Gearbox oil temperature	°C	M37	Average yaw power	kW
M13	Generator bearing temperature in driving	°C	M38	Maximum yaw power	kW
M14	Generator bearing temperature in non-driving	°C	M39	Angle, blade 1	°
M15	Maximum generator winding temperature	°C	M40	Angle, blade 2	°
M16	Temperature, pitch motor 1	°C	M41	Angle, blade 3	°
M17	Temperature, pitch motor 2	°C	M42	Maximum angle, blade 1	°
M18	Temperature, pitch motor 3	°C	M43	Maximum angle, blade 2	°
M19	Torque, pitch motor 1	Nm	M44	Maximum angle, blade 3	°
M20	Torque, pitch motor 2	Nm	M45	Generator average speed	rpm
M21	Torque, pitch motor 3	Nm	M46	Generator minimum speed	rpm
M22	Yaw converter temperature	°C	M47	Generator maximum speed	rpm
M23	Nacelle temperature	°C	M48	Control generator average speed	rpm
M24	Active power	kW	M49	Control generator minimum speed	rpm
M25	Reactive power	kVa	M50	Control generator maximum speed	rpm

monitoring system is alarmed (i.e., Fig. 18 (d)), the parameters such as M5,M6,M9,M10,M19-M21,M25,M31,M39,M40,M41,M45-M50, etc., show a higher degree of abnormality, while in the moments before the fault (i.e., Fig. 18 (b) and (c)), only the parameters M9,M31,M42-M44, M46 and M49 are abnormal. Therefore, the root cause of this fault can be located in the blade, inverter cabinet or generator, which can be emphasized in the subsequent maintenance process.

5.4. Discussions

Physics-guided explicit knowledge. For practical application of GNNs, “using a pre-defined graph for model training” [12] is a difficult task because the pre-defined graph do not exist in many industrial scenarios (e.g., constructing a digital twin model for an equipment). Therefore, it is a great concern that how to obtain the pre-defined graph explicitly using some rules. For the scenario of the digital twin model construction for wind turbines, this paper provides an idea to produce a pre-defined graph by defining a series of explicit knowledge describing interactions between variables from five aspects including mechanical

transmission relationships, temperature influence relationships, correlations, mutual information and causality, which also provides a solution of obtaining a pre-defined graph in the industrial scenario of constructing a digital twin model for an equipment.

Data-driven implicit knowledge and knowledge fusion. It should be noted that the explicit knowledge cannot fully and correctly express the relationships between nodes in the graph. To obtain a pre-defined graph with greater probability to match the real world scenarios, it is necessary to extract the implicit knowledge from data and fuse these two types of knowledge. To achieve this goal, this paper, based on the previous research [31], implement the knowledge fusion with the implicit knowledge as the core and the explicit knowledge with different weights as the supplement in a quasi-hard-attention manner. This method can be embedded in the process of explicit-implicit knowledge fusion in different scenarios to obtain a pre-defined graph required for GNNs training effectively.

Spatio-temporal feature learning and modeling accuracy. In this paper, the digital twin model construction for wind turbines is abstracted as the multivariate time series forecasting. Compared to traditional time-series neural networks limited by paying more attention to temporal dependencies without considering or only implicitly considering inter-variable dependencies, the time-series graph neural networks that simultaneously consider inter-variable dependencies and temporal dependencies has a superior prediction performance. Therefore, a high-precision digital twin model construction framework using the explicit-implicit fused graph obtained from the knowledge fusion module and the MTGNN [11] is proposed in this paper, and its modeling accuracy (RMSE: 0.0427 and R^2 : 0.9666) is compared with that of traditional time-series neural networks and other time-series graph neural networks, verifying the advantage on time series forecasting of time-series graph neural networks on one hand, and verifying the prediction performance enhancement of the fused explicit-implicit knowledge to the time-series graph neural networks on the other hand. In fact, this framework is highly generalizable, and the construction of high-precision digital twin models for different devices/systems (e.g., the coal pulverizer and the Lublin sugar factory in this paper) can be accomplished with the same idea.

Fault propagation analysis and root cause localization. In this paper, fault propagation analysis is achieved first by accurately describing the anomaly degree (the degree of deviation from the normal state) of each parameter at each moment, and then analyzing the propagation of the fault state among the parameters on a moment-by-moment basis. The study of fault propagation is valuable in both academic and practical aspects. On the one hand, this research makes the fault diagnosis change from the study of “what” to the study of “why” by investigating the propagation of fault modes in complex systems and further contributes to the discovery of fault propagation mechanism. On the other hand, when this research is combined with industrial practice, the factory operators will focus on both the fault-equipment (the equipment that is currently alarming) and the fault-source-equipment (the equipment that initially generates the fault mode), which will improve the level of systematic O&M within the plan and reduce the risk of unplanned downtime.

Applications on low-sampling-rate data and framework generalization ability. A large amount of multivariate monitoring data with low sampling rate, high proportion of normal conditions, and low cost exists in various scenarios in the real world, and it is a much-attended and difficult problem that how to effectively use these data for equipment fault warning, detection, localization, and diagnosis. To this end, this paper proposes an effective solution to handle such data and use it for equipment O&M, i.e., constructing a high-precision digital twin model under normal conditions, and realizing anomaly detection and root cause localization by tracking the deviation of the actual equipment’s state relative to the normal state. In the case study, this paper successfully reproduces the propagation of actual faults in the wind turbine, the coal pulverizer and the sugar factory system, and achieves

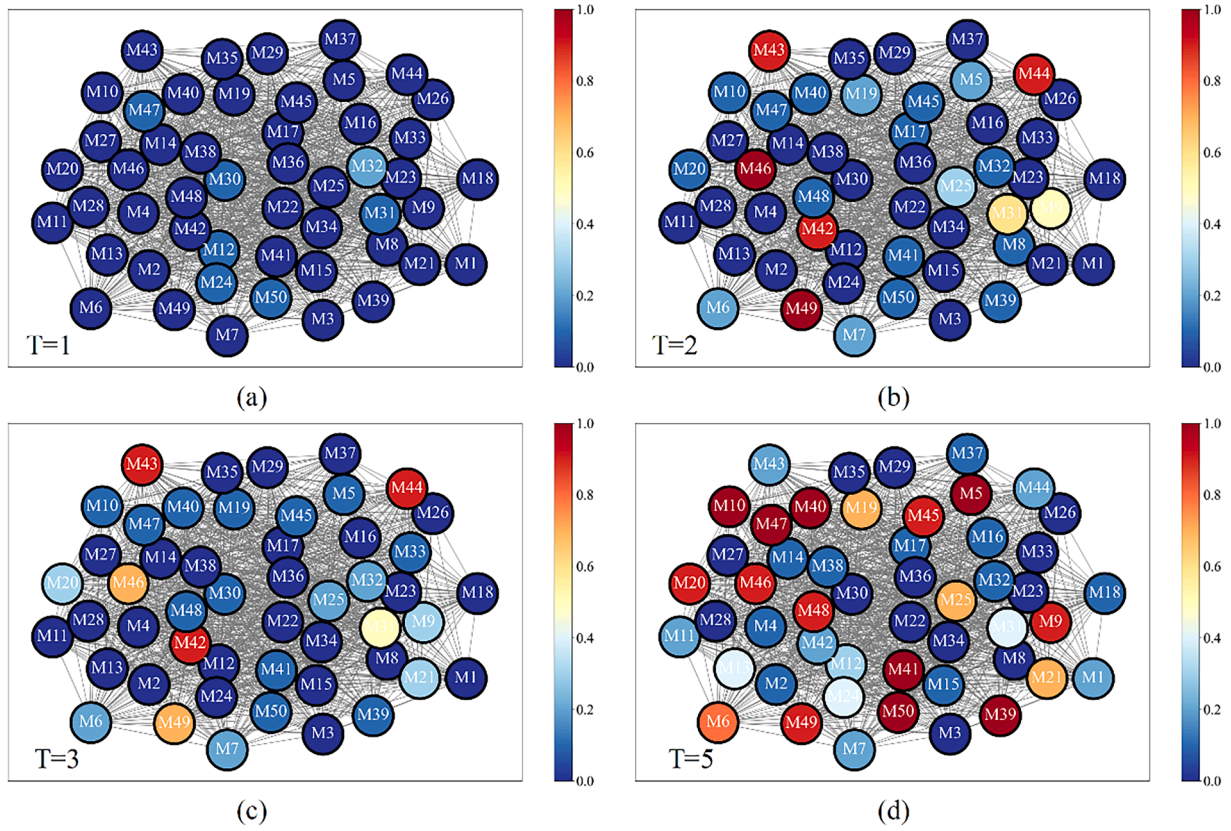


Fig. 18. Fault propagation analysis for Fault "Inverter Abnormality".

root cause localization through backtracking analysis. In summary, this study not only provides an idea on how to use low sampling rate monitoring data to achieve fault location, but also illustrates that the prediction deviations-based anomaly detection strategy, with its simplicity and reliability, can ensure the generalization of the proposed framework.

6. Conclusion

Motivated by the urgent demand to realize root cause localization for wind turbines using a large amount of monitoring data, this paper proposes a root cause localization framework for wind turbines based on explicit-implicit knowledge fusion and MTGNN by fully utilizing the powerful capability of GSL for inter-variable dependencies mining and the superior capability of time-series graph neural networks for multivariate time series forecasting. The explicit knowledge among variables for wind turbines is constructed based on five explicit rules including mechanical transmission relationships, temperature influence relationships, correlation, mutual information and causality, which represent the inter-variable dependencies in an explicit way. Based on the idea of GSL, the implicit knowledge learned from data and the explicit knowledge are weighted and fused in a quasi-hard-attention manner to obtain the graph adjacency matrix required for GNNs. Using the MTGNN framework, a high-precision digital twin model is trained based on the fused knowledge. Finally, the anomaly degree of a node's state is defined to accurately portray the propagation of the fault's state over time, after that the root cause localization is achieved by backtracking analysis. The proposed framework is validated using fault cases of wind turbines, a coal pulverizer and a sugar factory in the real world, and the results show that: i) the explicit knowledge defined by five explicit rules can partially describe the inter-variable dependencies in a wind turbine. ii) the fused graph learned by the GSL algorithm and the weighted fusion module using the quasi-hard-attention mechanism can more richly

express the inter-variable dependencies, and improve the modeling accuracy of MTGNN. iii) Based on the high-precision digital twin model for wind turbines and the defined anomaly degree for a node's state, the proposed framework can effectively describe the propagation of the fault's state and successfully realize the root cause localization by backtracking analysis. iv) the proposed root cause localization framework can be widely applied to different equipment to improve the O&M efficiency and reduce costs. In the future research, different early warning algorithms can be used to analyze the multivariate prediction deviations to enhance the early warning capability of the digital twin model.

CRedit authorship contribution statement

Chenlong Feng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chao Liu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Formal analysis. **Dongxiang Jiang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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